Visa Token Service Acquirers Guide

Technical Specifications

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Chapter 1

Visa Token Service Program Overview

The payments industry is evolving to support new payment form factors that require increased protection against counterfeit, account misuse, and other forms of fraud. Further protections are needed for card-not-present and hybrid transaction environments to help minimize unauthorized use of cardholder account data and prevent cross-channel fraud. Tokenization holds substantial promise to address these needs.

Tokens are surrogate values that replace the cardholder primary account number (PAN) in payment transactions during transaction processing. Payment processes that need to support payment token processing may include authorization, capture, clearing, and exception processing.

Visa has implemented network and processing-level changes to align with the *EMV Payment Tokenisation Specification – Technical Framework* (payment token framework) published by EMVCo. These changes support acquirer and merchant token implementations to provide enhanced data security for the cardholder primary account number (PAN) in payment transactions that addresses processing considerations and use of payment tokens to protect cardholder payment credentials. The standard describes the data elements and their intended use by the payment ecosystem in order to support the use of payment tokens in transactions.

Visa adopts the payment token standard. Acquirers and merchants that choose to implement the optional service must support the applicable changes for token processing. Acquirers must be prepared to send and receive the payment token in original transactions and in exception processing.

The Visa Token Service (VTS) also supports network 0003 (Interlink) messages.

Acquirers that have successfully completed applicable testing may implement the Visa Token Service.

Benefits of the Visa Token Service

Benefits of the Visa Token Service include:

- The cost of fraud can be reduced by minimizing card re-issuance.
- In the event of a merchant data breach, risk of subsequent fraud can be reduced.
- There is less risk in storing tokens on mobile devices, in cloud-based mobile applications, and online by e-commerce merchants because tokens do not carry the consumer's PAN information.
- Multiple tokens can be issued for a single primary account, with each tied to a specific mobile device or service.
- Tokens are based on existing International Organization for Standardization (ISO) standards, and therefore can be processed and routed by merchants, acquirers, and issuers, much like traditional card payments.

For More Information

Visa staff is available to assist acquirers during development and implementation of the Visa Token Service. For more information, acquirers should contact their client support representative.

Chapter 2

Visa Token Service Acquirer Participation Requirements

Acquirers must be connected to the VisaNet systems and complete the required Host system changes to support payment token data fields.

For acquirer participation requirements and PCR activation in the VisaNet Systems Tables, contact your regional client support representative.

A separate project is required to implement the Visa Token Service. Contact your regional client support representative to initiate a new project.

Acquirers that choose to participate in the Visa Token Service are required to register with Visa and comply with the participation requirements, systems, and processes. After successful registration, the token requestor is assigned a Token Requestor ID. The token requestor can then begin processing token requests in accordance with the specified requirements.

Acquirer General Implementation Requirements

Acquirers that choose to participate in the Visa Token Service must support certain Visa participation requirements.

These requirements include:

- Use the VisaNet standard and VisaNet Integrated Payment (V.I.P.) International Standards Organization (ISO) message formats, message types, and observe all rules for its use.
- Support Visa Token Service point-of-sale (POS) token payments processing for Visa transactions using V.I.P. message formats.
- Support near field communication (NFC), card-on-file/e-commerce, magnetic secure transmission (MST), and application-based e-commerce (host card emulation (HCE) and secure element (SE)) POS authorization and full financial messages for payment token processing.
- Support key fields and data requirements in V.I.P. authorization only and full financial messages, including original credit transactions (OCTs) and account funding transactions (AFTs) for certain token types, for payment token standards, issuance, and token processing.
- Support the use of tokens provisioned by the Visa Token Service for certain token types for contactless ATM transactions.
- Support key fields and data requirements in BASE II for payment token processing.
- Support the retain and return of the token in exception item processing.
- Recognize the token indicators that identify token account ranges in the Edit Package ARDEF Table and Routing Tables.
- Ensure all platforms are processing based on the processing and routing attributes defined for each nine-digit account range within the ARDEF and routing tables.
- Support the SMS Raw Data record V22201 that carries token data elements, if subscribing to SMS Raw Data.

Chapter 3

Visa Token Service Acquirer Payment Processing

Acquirers who participate in the Visa Token Service must be aware of the V.I.P. and BASE II message requirements and key fields necessary to support payment processing.

Visa Token Service Acquirer Transactions

Acquirers will process all transactions in the same manner as they do today, including authorization, capture, clearing, and exception processing. However, additional fields may be required to support the Visa Token Service.

Acquirers that choose to implement the Visa Token Service must support the payment token data elements from application-based, cloud-based, and mobile devices for:

- Near field communication (NFC)
- Card-on-file/E-Commerce
- E-commerce enabler

- Application-based e-commerce
  - Secure element (SE)
  - Host card emulation (HCE)
- Cloud-based magnetic secure transmission (MST)

Participating acquirers must be prepared to send and receive payment token data in original transactions and in exception items.

V.I.P. Payment Token Fields

Acquirers must continue to support the existing processing of VisaNet system messages, fields, and values, and be prepared to support the V.I.P. key fields for payment token transactions.

The fields are shown in this table.

V.I.P. Payment Fields for Visa Token Service

Field	Description
Field 2	Primary Account Number will contain the token
Field 14	Date, Expiration will contain the token expiration date
Field 22	Point-of-Service Entry Mode Code will contain the token presentment mode, depending on the type of transaction
Field 23	Card Sequence Number will contain the token sequence number that is forwarded from the terminal
Field 25	Point-of-Service Condition Code, with one of the following values depending on the type of POS transaction: • **00** (Normal transaction of this type) • **08** (Mail, telephone, recurring, advance, or installment order) • **51** (Address/CVV2/account verification without authorization; product eligibility inquiry without authorization; Mastercard POS account status inquiry) • **59** (E-commerce request by public network)
Field 35	Track 2 Data may carry token-specific data
Field 38	Authorization Identification Response contains the authorization code provided by the issuer

V.I.P. Payment Fields for Visa Token Service

Field	Description
Field 44.3	Additional Token Response Information contains a value used to identify transactions eligible for token services. It should only be sent in acquirer advices used for the clearing of an authorized transaction. **Note:** V.I.P. will not assign the value of **1** (Token Program) for the following messages: • 0100/0110 Preauthorization Request and Response • 0120/0130 Completion Advice and Response • 0220/0230 Preauthorization Completion and Response • 0400/0410 Preauthorization Reversal Request and Response • 0420/0430 Preauthorization Reversal Advice and Response • 0400/0410 Preauthorization Completion Reversal Request and Response • 0420/0430 Preauthorization Completion Reversal Advice and Response
Field 44.5	CVV/iCVV Results Code will be populated with the same value as the card authentication method (CAM) result code that will be returned in Field 44.8
Field 44.8	Card Authentication Results Code contains a Visa-defined code to indicate the online card authentication method (Online CAM) results
Field 44.10	CVV2 Result Code
Field 44.13	CAVV Results Code may contain the token verification result code
Field 44.15	Primary Account Number, Last Four Digits for Receipt will carry the last four digits of the cardholder PAN in the response to the acquirer
Field 45	Track 1 Data may carry token-specific data
Field 55, Usage 1	VSDC Chip Data, with Visa-generated chip data for token data elements in NFC token processing
Field 56	Customer Related Data in TLV format, Dataset ID 01—Payment Account Data, Tag 01—Payment Account Reference, with the payment account reference value
Field 60.2	Terminal Entry Capability
Field 60.8	Mail/Phone/Electronic Commerce and Payment Indicator, with the existing values of **05** (Secure electronic commerce transaction) or **07** (Non-authenticated security transaction) for card-on-file, e-commerce, or application-based e-commerce transactions from HCE and SE mobile devices. The acquirer should submit the value they received from the merchant
Field 62.23	Product ID will contain the product ID of the token PAN in the authorization response
Field 62.26	Account Status, that will identify the account range as regulated or non-regulated

V.I.P. Payment Fields for Visa Token Service

<table>
<tr><th>Field</th><th>Description</th></tr>
<tr><td>Field 63.6</td><td>Chargeback Reduction/BASE II Flags, position 4—Mail/Phone, Electronic Commerce and Payment Indicator, with the existing values 5 (Secure electronic commerce transaction) or 7 (Non-authenticated security transaction) for card-on-file, e-commerce, and application-based e-commerce transactions from HCE and SE mobile devices</td></tr>
<tr><td>Field 123, Usage 2</td><td>Verification & Token Data (TLV format), with these Dataset IDs:
Dataset ID 66—Verification Data
Dataset ID 67—Activation Verification Data
• Tag 08—Token Verification Result Code
– 1—TAVV cryptogram failed validation
– 2—TAVV cryptogram pass validation
– 3—DTVV or Visa-defined format cryptogram failed validation
– 4—DTVV or Visa-defined format cryptogram passed validation
Dataset ID 68—Token Data
• Tag 01—Token, that will carry the token
• Tag 02—Token Assurance Method, with a value that indicated the confidence level of the token to PAN/cardholder binding
• Tag 03—Token Requestor ID, that will carry the token requestor ID value
• Tag 04—Primary Account Number, Account Range, that will carry the first nine digits of the cardholder PAN
• Tag 05—Token Reference ID, that will contain the token reference ID
• Tag 06—Token Expiration Date, in yymm format
• Tag 07—Token Type, that will contain the token type value:
– 01—E-commerce/card-on-file
– 02—Secure element (SE)
– 03—Host card emulation (HCE), Cloud-based payment (CBP)
– 05—E-commerce enabler
– 06—Pseudo account</td></tr>
<tr><td>Field 125, Usage 2</td><td>Supporting Information (TLV format)</td></tr>
</table>

V.I.P. Payment Fields for Visa Token Service

Field	Description
Field 126.8	3-D Secure TAVV, Version and Authentication Action for payment tokens and token cryptograms present in e-commerce POS authorization and full financial messages, is used in conjunction with Field 126.9 for 3-D Secure (3DS) and non-3DS token transactions. For 3DS token transactions, the acquirer can: • Populate Field 126.8 with the TAVV • Populate Field 126.9 with the CAVV For non–3DS token transactions, the acquirer can: • Populate either Field 126.8 or Field 126.9 with the TAVV • There is no CAVV
Field 126.9	Usage 3: 3-D Secure CAVV, Revised Format, position 7, CAVV Version and Authentication Action, with the values in a Visa-defined format for payment tokens and token cryptograms
Field 126.10	CVV2 Authorization Request Data, will contain the Visa-generated value for dynamic token verification value (DTVV)
Field 126.13	POS Environment, will contain a value that indicates the type of merchant-initiated transaction
Field 126.20	3-D Secure Indicator

These existing fields may contain token-specific data:

- Payment tokens and token cryptograms that are present in NFC POS authorization and full financial messages will contain the token-specific data when forwarded from the terminal in Field 55, Usage 1—VSDC Chip Data or in Field 134—Visa Discretionary Data, Format 2, Expanded Format for third bitmap, depending on acquirer participation. Acquirers will be required to support the token-specific data in Field 55, Usage 1 or the third bitmap.
- Payment tokens and token cryptograms that are present in e-commerce POS authorization and full financial messages include Field 126.9, Usage 3, position 7. Acquirers can optionally send token-specific data in Field 126.9, Usage 3, position 7.

Visa Token Service Account Funding and Original Credit Transactions

Originators that send AFTs and OCTs with a token must include all the valid fields and values required for AFTs and OCTs, and may only send these transactions to recipient issuers that support AFTs, OCTs, and the Visa Token Service. All available token types are supported.

Note: Sender data is no longer required to be submitted on BASE II-originated OCTs that contain a token provisioned by the Visa Token Service. Sender data will continue to be required in OCTs with a token for acquirers and originators that currently submit these transactions as 0200 Full financial transactions.

Visa Token Service Contactless ATM and Manual Cash Transactions

Acquirers that participate in the Visa Token Service can submit contactless ATM and contactless manual cash disbursement transactions for secure element (SE), embedded SE, and host card emulation (HCE) token types for certain transactions.

These transactions are:

- ATM cash withdrawals
- ATM balance inquiry
- ATM account transfer
- Shared deposit
- Manual cash

Important:

Testing and activation are required to support contactless ATM transactions with a token. Contact your regional client support representative for additional information.

V.I.P. Processing for Contactless ATM and Cash Disbursement Token Transactions

Contactless ATM and contactless manual cash disbursement transactions that are submitted for token processing must comply with all the existing processing rules for Visa (network 0002) and Plus (network 0004) ATM transactions.

Encryption processing rules and requirements apply. Transactions must comply with quick Visa Smart Debit/Credit (qVSDC) processing rules.

These are V.I.P. key fields and values for contactless ATM and contactless manual cash disbursement token transactions:

- Field 2—Primary Account Number contains a valid token PAN

  Note: This field must contain the token that was processed in the authorization or full financial message in Field 2—Primary Account Number (PAN).

- Field 3—Processing Code, positions 1–2 must be:
  - **01** (Cash disbursement)
  - **30** (Balance inquiry)
  - **40** (Cardholder account transfer)

  or

  - **21** (Deposit Plus)
- Field 18—Merchant Type must be **6011** (Financial institutions–automated cash disbursement) or **6010** (Financial institutions, manual cash disbursements)
- Field 22—Point-of-Service Entry Mode Code must be **07** (Contactless device-read—originated using VSDC chip data rules; Online CAM authentication method; iCVV checking possible)

BASE II Processing for Contactless ATM and Cash Disbursement Token Transactions

There are key BASE II TCRs, fields, and values required for the Draft Data, TC x7 (Cash disbursement) for contactless ATM and contactless manual cash disbursement token transactions.

These TCRs, fields, and values are:

- TCR 0
  - Positions 5–20, Account Number, with a valid token PAN

    Note: This field will contain the token that was processed in the authorization message in Field 2—Primary Account Number (PAN).

  - Merchant Category Code (positions 133–136) must be **6011** (Financial institutions–automated cash disbursement) or **6010** (Financial institutions, manual cash disbursements)
  - POS Entry Mode (positions 162–163) must be **07** (Proximity payment using VSDC chip data rules)
- TCR 1—Additional Data
- TCR 4—Supplemental Financial Data
- TCR 4—Supplemental Financial and Promotion Data

- TCR 5—Payment Service Data
- TCR 7—Chip Card Transaction Data

Interlink Token Processing

In addition to token transaction processing on network 0002 (Visa), Visa also supports token processing on network 0003 (Interlink). This allows for token requestors and Token Service Providers to effectively implement functions within their own environment to support the payment token framework.

Interlink acquirers and merchants that have successfully completed testing may implement the Visa Token Service to support Interlink transactions. Acquirers and merchants have the option to participate in payment token processing for Interlink POS preauthorization and full financial messages. Acquirers must be prepared to send and receive the payment token in original Interlink transactions and exception processing.

Acquirers that choose to participate in Interlink for token processing must be aware of these requirements for Interlink transactions:

- Existing key fields
- Key fields and data requirements in Interlink POS preauthorization and full financial messages for payment token issuance and token processing
- Retain and return of the token in exception item processing for representments and representment reversals

Cryptogram Authentication for VTS

To maintain the security of Visa chip-based payments, CVN 4x authentication cryptograms are used. Acquirers must ensure that their contact chip terminals and contactless Visa Contactless readers do not reject secure element (SE) mobile devices personalized for issuer application data formats, as defined for CVN 4x.

CVN 4x uses a new issuer application data (IAD) format that allows for differing cryptographic methods to be used to generate the application cryptograms. The IAD format 4 is supported in TLV Field 55, Usage 1—VSDC Chip Data, Tag 9F10—Issuer Application Data.

Dynamic Token Verification

The dynamic token verification value (DTVV) is a simplified cryptogram for use with a Visa Token Service based e-commerce transaction.

E-commerce token transactions require the use of a token authentication verification value (TAVV) with the token as part of the capabilities provided by the Visa Token Service. DTVV provides an alternative domain control for merchants using the Visa Checkout Open Platform (VCOP) and for card-on-file token requestors.

The DTVV cryptogram is used for token-based transactions, while PAN-based transactions continue to be processed with CVV2 cryptograms. The DTVV cryptogram is a three-digit Visa generated value that will be carried in Field 126.10—CVV2 Authorization Request Data from the acquirer.

Once the DTVV cryptogram is processed, V.I.P. will validate the DTVV and send the request message to the issuer. The DTVV value in Field 126.10 will not be sent to the issuer in the request message.

The DTVV result code will be carried in Field 44.10—CVV2 Result Code.

V.I.P. will interpret the cryptogram in an authorization or full financial message as DTVV instead of CVV2 when the message contains a token-based e-commerce or card-on-file transaction.

Note:

DTVV is currently only available to U.S. acquirers and issuers.

Token Transactions with 3-D Secure

The Visa Token Service supports 3-D secure (3DS) validation for token-based card-on-file, e-commerce, and application-based e-commerce transactions.

Cardholder authentication can be performed with tokens using the these cryptograms:

- Cardholder authentication verification value (CAVV) and token authentication verification value (TAVV), which can be carried together
- Visa defined format cryptogram

All acquirers that participate in the Visa Token Service and 3DS must support the CAVV and TAVV cryptograms for token-based transactions with 3DS.

The two separate cryptograms are supported in the authorization message, the TAVV token cryptogram for token validation, and the 3DS CAVV cryptogram for cardholder authentication.

Acquirers must submit the TAVV cryptogram data in Field 126.8—Transaction ID (XID) in combination with the 3DS cryptogram data in Field 126.9—Usage 3: 3-D Secure CAVV, Revised Format for token-based card-on-file, e-commerce, and application-based e-commerce transactions with 3DS 1.0.2, 3DS 2.0, or its subsequent versions.

TAVV token cryptogram data can be carried in combination with the 3DS CAVV cardholder authentication cryptogram. Additionally, the Visa-defined format cryptogram enables validation of both the token cryptogram and the 3DS CAVV cardholder authentication cryptogram.

Acquirers must be capable of receiving the token verification results in one of the these fields:

- Field 44.13—CAVV Results Code
- TLV Field 123, Usage 2, Dataset ID 67—Activation Verification Data, Tag 08—Token Verification Result Code

BASE II Payment Token Fields

Acquirers will send the token in BASE II Draft Data, TCR 0, positions 5–20, Account Number for transactions that were processed using the Visa Token Service. BASE II will replace the token with the cardholder PAN prior to forwarding the transaction to the issuer.

The transaction will be returned to the acquirer with return reason code **9E** (Token to PAN relationship cannot be located) if a token relationship to the cardholder PAN cannot be located. Items returned with return reason code **9E** will be settled at the acquiring identifier level instead of the CIB level.

Acquirers must be aware that the validation code will be based on the token and not on the cardholder PAN. Acquirers must continue to support the existing processing of VisaNet System messages, fields, and values, and be prepared to support these BASE II key fields for payment token transactions:

- Returned Credit, TC 01, TCR 9, positions 5–10, Destination Identifier, and positions 11–16, Source Identifier, with the submitting acquiring identifier
- Returned Debit, TC 02, TCR 9, positions 5–10, Destination Identifier, and positions 11–16, Source Identifier, with the submitting acquiring identifier
- Draft Data, TC x5, x6, and x7, TCR 0, positions 5–20, Account Number will contain the token
- Draft Data, TC x5, x6, and x7, TCR 0, positions 152–157, Authorization Code
- Draft Data, TC x5, x6, and x7, TCR 0, position 158, POS Terminal Capability
- Draft Data, TC x5, x6, and x7, TCR 0, positions 162–163, POS Entry Mode
- Draft Data, TC x5, x6, and x7, TCR 1, position 5, Business Format Code
- Draft Data, TC x5, x6, and x7, TCR 1, positions 6–7, Token Assurance Method
- Draft Data, TC x5 and TC x6, TCR 1, position 116, Mail/Phone/Electronic Commerce and Payment Indicator
- Draft Data, TC x5, x6, and x7, TCR 1, position 122, Additional Token Response Information
- Draft Data, TC x5 and TC x6, TCR 1, position 168, POS Environment
- Draft Data, TC x5, x6, and x7, TCR 4—Supplemental Financial Data, positions 120–148, Payment Account Reference

- Draft Data, TC x5, x6, and x7, TCR 4—Supplemental Financial Data, positions 149–159, Token Requestor ID
- Draft Data, TC x5, x6, and x7, TCR 4—Supplemental Financial and Promotion Data, positions 112–140, Payment Account Reference
- Draft Data, TC x5, x6, and x7, TCR 4—Supplemental Financial and Promotion Data, positions 141–151, Token Requestor ID
- Draft Data, TC x5, x6, and x7, TCR 7—Chip Card Transaction Data
- Multipurpose Messages, TC 33
- Text Messages, TC 50
- Retrieval Request Record, TC 52, TCR 0, positions 5–20, Account Number

  Important:

  Acquirers must include the authorization code from the authorization response message in the BASE II clearing transaction. The authorization code is carried in the BASE II draft data, TCR 0, positions 152–157. Transactions must be authorized prior to submitting the transaction to BASE II. Token transactions that are not authorized will be returned with the existing return reason code 17 (The authorization code is invalid).

When an endpoint sends the Additional Token Response Information field with an invalid value for the type of transaction sent, BASE II will replace the value with a space and continue processing the transaction.

Near Field Communication (NFC) Token Processing

Once the acquirer has obtained the token, near field communication (NFC) processing allows acquirers and their merchants to process POS authorization or full financial request and response transactions using the token instead of the cardholder PAN.

The NFC-capable terminal will capture the token data when the device is tapped against the reader device. Acquirers will send the token data and other fields and values in V.I.P. POS Visa Contactless authorization or full financial request messages.

V.I.P. Token Processing for NFC

Acquirers must support key fields for POS authorization or full financial request and response messages for NFC token processing.

Key Field Impacts NFC Token Processing

Key Fields	NFC Token Processing for POS Request Message	NFC Token Processing for POS Response Message
Field 2—Primary Account Number	✓	✓
Field 14—Date, Expiration	✓	
Field 22—Point-of-Service Entry Mode Code	✓	
Field 23—Card Sequence Number	✓	
Field 25—Point-of-Service Condition Code	✓	✓
Field 35—Track 2 Data	✓	
Field 38—Authorization Identification Response		✓
Field 44.5—CVV/iCVV Results Code		✓
Field 44.8—Card Authentication Results Code		✓
Field 44.15—Primary Account Number, Last Four Digits for Receipt		✓
Field 45—Track 1 Data	✓	
Field 55—Integrated Circuit Card (ICC)-Related Data	✓	

Field 22 Values for NFC Token Processing

Field 22—Point-of-Service Entry Mode Code is an existing field that must include the appropriate POS entry mode code for near field communication (NFC) token processing of POS contactless transactions.

For NFC processing, acquirers that choose to implement the Visa Token Service must submit the appropriate POS entry mode code in Field 22 of these V.I.P. messages.

- **07** (Contactless device-read-originated using VSDC chip data rules; Online CAM authentication method; iCVV checking possible)
- **91** (Contactless device-read-originated using magnetic stripe data rules; dCVV checking is possible; Online CAM checking is possible for MSD CVN 17 only)

This table shows the V.I.P. Systems messages required to support Field 22 for token processing.

V.I.P. Systems Message Impacts for Field 22

Message Type	Description	V.I.P.	
		Auth Only	Full Service
0100/0120	Authorization Request and STIP Advice	✓	✓
0200/0220	Full Financial Request, Acquirer Advice, STIP Advice, and BASE II Advice		✓
0220	Adjustment Advice and STIP Advice		✓
0400/0420	Reversal, Partial Reversal, and Reversal Advice	✓	✓
0400/0420	Financial Reversal, Acquirer Advice, Issuer STIP Advice, and Issuer Switch Advice		✓

Field 25 Values for NFC Token Processing

Field 25—Point-of-Service Condition Code is an existing field that must include the appropriate value for near field communication (NFC) token processing of POS contactless transactions.

For NFC processing, acquirers that choose to implement the Visa Token Service must submit the appropriate point-of-service condition code in Field 25 of V.I.P. messages with the value of **00** (Normal transaction of this type).

BASE II NFC Token Processing

Acquirers that choose to implement the Visa Token Service must confirm successful processing of clearing drafts and exception items that will include token data.

This table shows the key token data fields for the BASE II Draft Data, TCR 0 record.

TCR 0

Position	Length	Format	Field Name	Description
5–20	16	UN	Account Number	This field contains the cardholder PAN. For token transactions, this field will contain the token.
152–157	6	AN	Authorization Code	This field contains the authorization code. A token transaction must be authorized and this field must contain the same value as the authorization request.
162–163	2	AN	POS Entry Mode	This field contains the POS entry mode and should contain the same POS entry mode value as the authorization request. For NFC transactions, this field should contain one of these values: • **07** (Contactless device-read originated using VSDC chip data rules; Online CAM is the authentication method; iCVV checking is possible) • **91** (Contactless device-read originated using magnetic stripe data rules; Online CAM checking is possible; dCVV or iCVV checking is possible)

This table shows the key token data fields for the BASE II Draft Data, TCR 1 record.

TCR 1—Additional Data

Position	Length	Format	Field Name	Description
5	1	AN	Business Format Code	This field is reserved for future use. This field contains a space.
6–7	2	AN	Token Assurance Method	A value that allows the Token Service provider to indicate the trust level of the Token to PAN/cardholder. It is determined as a result of the type of identification and verification (ID & V) performed and the entity that performed it. If not used, this field should be space filled.

Card-On-File or E-Commerce Token Processing

Once the acquirer has obtained the token, acquirer card-on-file token processing allows acquirers and their merchants to process the transaction using the token instead of the cardholder PAN.

The acquirer will use the token instead of the cardholder PAN for transactions such as card-not-present transactions when POS authorizations are initiated and when clearing transaction processing takes place. Acquirers will use the token data and other fields and values to create and send in POS authorization or full financial request messages.

Acquirers will be required to support payment tokens and token cryptograms in Field 126.9—Usage 3: 3-D Secure CAVV, Revised Format, position 7 for card-on-file/e-commerce POS authorization and full financial messages originating from both secure element (SE) and host card emulation (HCE) mobile devices.

V.I.P. Card-on-File/E-Commerce Token Processing

Acquirers must use key V.I.P. fields and values for card-on-file token processing.

Key Fields for Card-On-File/E-Commerce Token Processing

Key Fields	Card-on-file Token Processing for POS Request Message	Card-on-file Token Processing for POS Response Message
Field 2—Primary Account Number	✓	✓
Field 14—Date, Expiration	✓	
Field 22—Point-of-Service Entry Mode Code	✓	
Field 25—Point-of-Service Condition Code	✓	✓
Field 44.13—CAVV Results Code		✓
Field 44.15—Primary Account Number, Last Four Digits for Receipt		✓
Field 60.8—Mail/Phone/Electronic Commerce and Payment Indicator	✓	

Key Fields for Card-On-File/E-Commerce Token Processing

Key Fields	Card-on-file Token Processing for POS Request Message	Card-on-file Token Processing for POS Response Message
Field 63.6—Chargeback Reduction/ BASE II Flags	✓	
Field 126.9—CAVV Data	✓	

Field 22 Values for Card-on-file or E-Commerce Token Processing

Field 22—Point-of-Service Entry Mode Code is an existing field that must include the appropriate POS entry mode code for card-on-file token processing of e-commerce or card-not-present POS transactions.

For card-on-file processing, acquirers that choose to implement the Visa Token Service must submit the appropriate POS entry mode code in Field 22 of V.I.P. messages:

- **01** (Manual (key entry))
- **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file)

This table shows the V.I.P. Systems messages required to support Field 22 for token processing.

V.I.P. Systems Message Impacts for Field 22

Message Type	Description	V.I.P.	
		Auth Only	Full Service
0100/0120	Authorization Request and STIP Advice	✓	✓
0200/0220	Full Financial Request, Acquirer Advice, STIP Advice, and BASE II Advice		✓
0220	Adjustment Advice and STIP Advice		✓
0400/0420	Reversal, Partial Reversal, and Reversal Advice	✓	✓
0400/0420	Financial Reversal, Acquirer Advice, Issuer STIP Advice, and Issuer Switch Advice		✓

Field 25 Values for Card-on-File or E-Commerce Token Processing

Field 25—Point-of-Service Condition Code is an existing field that must include the appropriate value for card-on-file token processing of POS transactions.

For card-on-file processing, acquirers that choose to implement the Visa Token Service must submit the appropriate point-of-service condition code in Field 25 of V.I.P. messages:

- **08** (Mail, telephone, recurring, advance, or installment order)
- **59** (E-commerce request by public network)

Field 60.8 Values for Card-on-file or E-Commerce Token Processing

Field 60.8—Mail/Phone/Electronic Commerce and Payment Indicator is an existing field that must include the appropriate value for card-on-file token processing of POS transactions.

For card-on-file processing, acquirers that choose to implement the Visa Token Service must submit the appropriate unmodified MOTO/ECI value in Field 60.8 of V.I.P. messages:

- **05** (Secure electronic commerce transaction)
- **07** (Non-authenticated security transaction)

BASE II Card-on-File or E-Commerce Token Processing

Acquirers that choose to implement the Visa Token Service must confirm successful processing of clearing drafts and exception items that will include token data.

This table shows the key token data fields for the BASE II Draft Data, TCR 0 record.

TCR 0

Position	Length	Format	Field Name	Description
5–20	16	UN	Account Number	This field contains the cardholder PAN. For token transactions, this field will contain the token.
152–157	6	AN	Authorization Code	This field contains the authori-zation code. A token transaction must be authorized and this field must contain the same value as the authorization request.
162–163	2	AN	POS Entry Mode	This field contains the POS entry mode and should contain the same POS entry mode value as the authorization request. For card-on-file transactions, this field should contain one of these values: • **01** (Card-on-file (manual key entry)) or • **10** (Credential on file)

This table shows the key token data fields for the BASE II Draft Data, TCR 1 record.

TCR 1—Additional Data

Position	Length	Format	Field Name	Description
5	1	AN	Business Format Code	This field is reserved for future use. This field contains a space.
6–7	2	AN	Token Assurance Method	A value that allows the Token Service provider to indicate the trust level of the Token to PAN/ cardholder. It is determined as a result of the type of identification and verification (ID & V) performed and the entity that performed it. If not used, this field should be space filled.

E-Commerce Enabler Token Processing

E-commerce enabler supports tokens where the token may be used across multiple e-commerce merchants.

Once the acquirer has obtained the token, acquirer card-on-file token processing allows acquirers and their merchants to process the transaction using the token instead of the cardholder PAN. The acquirer will use the token instead of the cardholder PAN when POS authorizations are initiated, and when clearing transaction processing takes place. Acquirers will use the token data and other fields and values to create and send in POS authorization or full financial request messages.

V.I.P. Processing for E-Commerce Enabler Token Processing

Acquirers use key V.I.P. fields and values for e-commerce enabler token processing.

Key Fields for E-Commerce Enabler Token Processing

Key Fields	COF Token Processing for POS Request Message	COF Token Processing for POS Response Message
Field 2—Primary Account Number	✓	✓
Field 14—Date, Expiration	✓	
Field 22—Point-of-Service Entry Mode Code	✓	
Field 25—Point-of-Service Condition Code	✓	✓
Field 44.13—CAVV Results Code		✓
Field 44.15—Primary Account Number, Last Four Digits for Receipt		✓
Field 60.8—Mail/Phone/Electronic Commerce and Payment Indicator	✓	
Field 63.6—Chargeback Reduction/ BASE II Flags	✓	
Field 126.9—CAVV Data	✓	

Field 22 Values for E-Commerce Enabler Token Processing

Field 22—Point-of-Service Entry Mode Code is an existing field that must include the appropriate POS entry mode code for e-commerce enabler token processing of e-commerce or card-not-present POS transactions.

For e-commerce enabler processing, acquirers that choose to implement the Visa Token Service must submit the appropriate POS entry mode code in Field 22 of V.I.P. messages:

- **01** (Manual (key entry))
- **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file)

This table shows the V.I.P. Systems messages required to support Field 22 for token processing.

V.I.P. Systems Message Impacts for Field 22

Message Type	Description	V.I.P.	
		Auth Only	Full Service
0100/0120	Authorization Request and STIP Advice	✓	✓
0200/0220	Full Financial Request, Acquirer Advice, STIP Advice, and BASE II Advice		✓
0220	Adjustment Advice and STIP Advice		✓
0400/0420	Reversal, Partial Reversal, and Reversal Advice	✓	✓
0400/0420	Financial Reversal, Acquirer Advice, Issuer STIP Advice, and Issuer Switch Advice		✓

Field 25 Values for E-Commerce Enabler Token Processing

Field 25—Point-of-Service Condition Code is an existing field that must include the appropriate value for e-commerce enabler token processing of POS transactions.

For e-commerce enabler processing, acquirers that choose to implement the Visa Token Service must submit the appropriate point-of-service condition code in Field 25 of V.I.P. messages:

- **08** (Mail, telephone, recurring, advance, or installment order)
- **59** (E-commerce request by public network)

BASE II E-Commerce Enabler Token Processing

Acquirers that choose to implement the Visa Token Service must confirm successful processing of clearing drafts and exception items that will include token data.

This table shows the key token data fields for the BASE II Draft Data, TCR 0 record.

TCR 0

Position	Length	Format	Field Name	Description
5–20	16	UN	Account Number	This field contains the cardholder PAN. For token transactions, this field will contain the token.
152–157	6	AN	Authorization Code	This field contains the authorization code. A token transaction must be authorized and this field must contain the same value as the authorization request.
162–163	2	AN	POS Entry Mode	This field contains the POS entry mode and should contain the same POS entry mode value as the authorization request. For e-commerce enabler transactions, this field should contain one of these values: • **01** (Card-on-file (manual key entry)) or • **10** (Credential on file)

This table shows the key token data fields the BASE II Draft Data, TCR 1 record.

TCR 1—Additional Data

Position	Length	Format	Field Name	Description
5	1	AN	Business Format Code	This field is reserved for future use. This field contains a space.
6–7	2	AN	Token Assurance Method	A value that allows the Token Service provider to indicate the trust level of the Token to PAN/ cardholder. It is determined as a result of the type of identification and verification (ID & V) performed and the entity that performed it. If not used, this field should be space filled.

Application-Based E-Commerce (Secure Element and Host Card Emulation) Token Processing

The Visa Token Service supports payment token processing for transactions originating from both secure element (SE) and host card emulation (HCE) mobile devices.

Once the acquirer has obtained the token, application-based e-commerce processing allows acquirers and their merchant to process POS authorization or full financial request and response transactions using the token instead of the cardholder PAN.

> **Note:**
>
> Acquirers will be required to support payment tokens and token cryptograms in Field 126.9—Usage 3–3-D Secure CAVV, Revised Format, position 7 for e-commerce POS authorization and full financial messages originating from both secure element (SE) and host card emulation (HCE) mobile devices.

V.I.P. Processing for Application-Based E-Commerce Token Processing

This table provides the key fields that acquirers must support for application-based e-commerce token processing.

Key Fields for Application-Based E-Commerce Token Processing

Key Fields	Application-Based E-Commerce Processing for POS Request Message	Application-Based E-Commerce Processing for POS Response Message
Field 2—Primary Account Number	✓	✓
Field 14—Date, Expiration	✓	
Field 22—Point-of-Service Entry Mode Code	✓	
Field 25—Point-of-Service Condition Code	✓	
Field 44.13—CAVV Results Code		✓
Field 44.15—Primary Account Number, Last Four Digits for Receipt		✓

Key Fields for Application-Based E-Commerce Token Processing

Key Fields	Application-Based E-Commerce Processing for POS Request Message	Application-Based E-Commerce Processing for POS Response Message
Field 60.8—Mail/Phone/ Electronic Commerce and Payment Indicator	✓	
Field 63.6—Chargeback Reduction/BASE II Flags	✓	
Field 126.9–Usage 3: 3-D Secure CAVV, Revised Format	✓	

Note:

Acquirers must be aware that application-based e-commerce transactions from SE mobile devices submitted with the MOTO/ECI value of **05** (Secure electronic commerce transaction) in Field 60.8 in V.I.P. Authorization request messages may be changed to **07** (Non–authenticated security transaction) by Visa. Acquirers that choose to receive the MOTO/ECI value in Field 60.8 of the 0110 Authorization response must be prepared to submit the value in the BASE II original transaction.

Application-based e-commerce transactions from SE mobile devices submitted with the MOTO/ECI value of **5** (Secure electronic commerce transaction) in Field 63.6 in V.I.P. full financial request messages and in the BASE II TC 05 or TC 06 transactions (TCR 1, position 116) may be changed to the value of **7** (Non-authenticated security transaction) by Visa.

Field 22 Values for Application-Based E-Commerce Token Processing

Field 22—Point-of-Service Entry Mode Code is an existing field that must include the appropriate POS entry mode code for application-based token processing.

For application-based e-commerce processing, acquirers that choose to implement the Visa Token Service must submit the appropriate POS entry mode code in Field 22 of V.I.P. messages:

- Secure Element
  - **01** (Manual (key entry))
  - **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file)
- Host Card Emulation
  - **01** (Manual (key entry))
  - **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file)

  Note:

  Acquirers will be required to support payment tokens and token cryptograms in Field 126.9—Usage 3: 3-D Secure CAVV, Revised Format, position 7 for application-based e-commerce POS authorization and full financial messages originating from both secure element (SE) and host card emulation (HCE) mobile devices.

This table shows the V.I.P. Systems messages required to support Field 22 for token processing.

V.I.P. Systems Message Impacts for Field 22

Message Type	Description	V.I.P.	
		Auth Only	**Full Service**
0100/0120	Authorization Request and STIP Advice	✓	✓
0200/0220	Full Financial Request, Acquirer Advice, STIP Advice, and BASE II Advice		✓
0220	Adjustment Advice and STIP Advice		✓
0400/0420	Reversal, Partial Reversal, and Reversal Advice	✓	✓
0400/0420	Financial Reversal, Acquirer Advice, Issuer STIP Advice, and Issuer Switch Advice		✓

Field 25 Values for Application-Based E-Commerce Token Processing

Field 25—Point-of-Service Condition Code is an existing field that must include the appropriate value for application-based e-commerce token processing of POS transactions.

For application-based e-commerce processing, acquirers that choose to implement the Visa Token Service must submit the appropriate point-of-service condition code in Field 25 of V.I.P. messages:

- **08** (Mail, telephone, recurring, advance, or installment order)
- **59** (E-commerce request by public network)

BASE II Application-Based E-Commerce Token Processing

Acquirers that choose to implement the Visa Token Service must confirm successful processing of clearing drafts and exception items that will include token data.

This table shows the key token data fields for the BASE II Draft Data, TCR 0 record.

TCR 0

Position	Length	Format	Field Name	Description
5–20	16	UN	Account Number	This field contains the cardholder PAN. For token transactions, this field will contain the token.
152–157	6	AN	Authorization Code	This field contains the authorization code. A token transaction must be authorized and this field must contain the same value as the authorization request.
162–163	2	AN	POS Entry Mode	This field contains the POS entry mode and should contain the same POS entry mode value as the authorization request. For application-based token transactions, this field should contain the following values: • **01** (Manual key entry) • **10** (Credential on file)

This table shows the key token data fields for the BASE II Draft Data, TCR 1 record.

TCR 1—Additional Data

Position	Length	Format	Field Name	Description
5	1	AN	Business Format Code	This field is reserved for future use. This field contains a space.
6–7	2	AN	Token Assurance Method	A value that allows the Token Service provider to indicate the trust level of the Token to PAN/ cardholder. It is determined as a result of the type of identification and verification (ID & V) performed and the entity that performed it. If not used, this field should be space filled.

Cloud-Based Magnetic Secure Transmission Token Processing

The magnetic secure transmission (MST) simulates a physical magnetic stripe transaction sent from a mobile phone or other device.

The magnetic stripe data, or track data, is transmitted through a contactless medium to any standard magnetic stripe reader, and requires no change to existing merchant POS or acquirer systems. The Visa Token Service supports processing of payment token transactions based on MST. A mobile device provisioned with a cloud-based payment token will be capable of supporting payment transactions using Visa Contactless near field communication (NFC) and MST.

Once the acquirer has obtained the token, cloud-based MST processing allows acquirers and their merchants to process POS authorization or full financial request and response transactions using the token instead of the cardholder PAN.

V.I.P. Token Processing for MST

Acquirers must support key fields for MST token processing.

Key Fields for Cloud-based MST Token Processing

Key Fields	Cloud-Based MST Processing for POS Request Message	Cloud-Based MST Processing for POS Response Message
Field 2—Primary Account Number	✓	✓
Field 14—Date, Expiration	✓	

Key Fields for Cloud-based MST Token Processing

Key Fields	Cloud-Based MST Processing for POS Request Message	Cloud-Based MST Processing for POS Response Message
Field 22—Point-of-Service Entry Mode Code	✓	
Field 25—Point-of-Service Condition Code	✓	✓
Field 35—Track 2 Data	✓	
Field 44.5—CVV/iCVV Results Code		✓
Field 44.15—Primary Account Number, Last Four Digits for Receipt		✓
Field 45—Track 1 Data	✓	

Field 22 Values for Cloud-Based MST Token Processing

Field 22—Point-of-Service Entry Mode Code is an existing field that must include the appropriate POS entry mode code for cloud-based MST token processing.

For cloud-based MST processing, acquirers that choose to implement the Visa Token Service must submit the appropriate POS entry mode code in Field 22 of V.I.P. messages with the value of **90** (Magnetic stripe read and exact content of Track 1 or Track 2 included (CVV check possible)).

This table shows the V.I.P. Systems messages required to support Field 22 for token processing.

V.I.P. Systems Message Impacts for Field 22

Message Type	Description	V.I.P.	
		Auth Only	Full Service
0100/0120	Authorization Request and STIP Advice	✓	✓
0200/0220	Full Financial Request, Acquirer Advice, STIP Advice, and BASE II Advice		✓
0220	Adjustment Advice and STIP Advice		✓
0400/0420	Reversal, Partial Reversal, and Reversal Advice	✓	✓
0400/0420	Financial Reversal, Acquirer Advice, Issuer STIP Advice, and Issuer Switch Advice		✓

Field 25 Values for Cloud-Based MST Token Processing

Field 25—Point-of-Service Condition Code is an existing field that must include the appropriate value for cloud-based MST token processing of POS transactions.

For cloud-based MST token processing, acquirers that choose to implement the Visa Token Service must submit the appropriate point-of-service condition code in Field 25 of V.I.P. messages with the value of **00** (Normal transaction of this type).

Additional System and Processing Considerations for the Visa Token Service

There are additional processing requirements, reports, and files that may be required for an acquirer to successfully support the Visa Token Service.

Commercial Card Token Processing

Acquirer processors, acquirers, and merchants that are directly connected to Visa and process transactions containing a token must support the fields that carry token data elements for commercial card payment token processing originating from secure element (SE) and host card emulation (HCE) mobile devices and magnetic secure transmission (MST).

Endpoints must be capable of sending, receiving, and processing token data elements for near field communication (NFC) and application-based e-commerce for SE, HCE, and MST POS authorization messages, full financial messages, and clearing transactions. Payment token processing for point-of-sale (POS) commercial transactions are applicable to secure element (SE) and host card emulation (HCE) mobile devices, magnetic secure transmission (MST), application-base e-commerce, e-commerce, and card-on-file.

The processing that applies to BASE II TC 50 transactions will also be applicable to commercial NFC, MST, card-on-file, e-commerce, and application-based e-commerce transactions. Commercial card acquirers will continue to send market-specific additional data in TC 50 transactions to commercial card issuers.

> **Note:**
>
> Support for tokenized Visa Purchasing cards in the Europe region will be available at a future date. Contact your regional client support representative for further information.
>
> **Important:**
>
> **Acquirers must populate the Destination BIN field in the TC 50 with the value from the Issuer BIN field, positions 25–30 from the ARDEF extract file table.**

Acquirers must not use the first six digits of the primary account number (PAN) or the token to determine the destination BIN.

Payment Account Reference Token Processing

The payment account reference is a data element in authorization messages and clearing transactions, and is associated with a cardholder's primary account number (PAN).

When the PAN is tokenized, the payment account reference will then be associated with all the payment tokens for the cardholder PAN. Therefore, the payment account reference allows acquirers to link transactions containing PANs that have been tokenized.

Note:

The payment account reference applies to both tokenized and non-tokenized PANs.

Acquirers must be aware that payment account reference values may be available for tokenized PANs. The payment account reference (PAR) number will be populated on all transactions related to the underlying PAN regardless of whether the transaction occurred with the PAN or related tokens.

Acquirers must be able to process a transaction that includes the payment account reference and distribute the value to the merchant.

Note:

For more information regarding the payment account reference, contact your regional client support representative.

Payment Account Reference V.I.P. Processing for VTS

The payment account reference is located in TLV Field 56—Customer Related Data, Dataset ID 01—Account Data, Tag 01—Payment Account Reference.

Acquirers must support TLV Field 56 in these transaction messages that are sent from V.I.P. to acquirers:

- Authorization and full financial request and response messages
- Exception item messages
- Acquirer advice, administrative, and status messages
- Request for copy messages

Acquirers that wish to inquire on a payment account reference value associated with a token (or PAN) may submit a V.I.P. 0300 Payment Account Reference inquiry message and must be prepared to receive the payment account reference in the V.I.P. 0310 Payment Account

Reference Inquiry response message. Acquirers must provide this information in the V.I.P. 0300 inquiry message:

- One of these values is present:
  - Field 02—Primary Account Number, or
  - Field 123, Usage 2—Verification & Token Data (TLV format), Dataset ID 68—Token Data, Tag 01—Token
- Field 91—File Update Code must be **5** (Inquire)
- Field 101—File Name must be **PAR** (Payment account reference)

The 0310 Payment Account Reference Inquiry response message will provide the payment account reference in Tag 01 of TLV Field 56, Dataset ID 01 and the payment account reference creation date in Tag 02—Payment Account Reference Creation Date.

Acquirers that choose to receive TLV Field 56 from Visa must be aware that they will receive values only in message for which a payment account reference is available.

Important:

After acquirers implement support for Field 56 in TLV format, they must be able to receive any dataset IDs and tags defined for this field in any order, including those that they do not recognize or expect. Endpoints may receive multiple datasets in the same Field 56. Not all defined dataset IDs are documented in the V.I.P. technical documentation. Endpoints must ignore any dataset IDs or tags they do not recognize, and continue to process the field.

Payment Account Reference BASE II Processing for VTS

The payment account reference is located in certain BASE II records.

These records are:

- Draft Data, TCx5, x6, and x7
  - TCR 4—Supplemental Financial Data, positions 120–148, Payment Account Reference
  - TCR 4—Supplemental Financial and Promotion Data, positions 112–140, Payment Account Reference
- TC 33—Authorization Records
  - Authorization and Incremental Authorization (POS) Record, TCR 4, positions 5–33, Payment Account Reference
  - Authorization Full and Partial Reversal (PSR) Record, TCR 4, positions 5–33, Payment Account Reference
- TC 33.A—Acquirer Capture File (Enhanced), CP01, TCR 8, positions 79–107, Payment Account Reference

- TC 33.B—Merchant Capture Files, CP51, TCR 8, positions 81–109
- TC 40—Fraud Advice Transaction, TCR 2, positions 88–116, Payment Account Reference

  Note:

  The payment account reference will only be present in a BASE II TC record if submitted by the acquirer or issuer. Visa will not generate the payment account reference on any BASE II TC record.

  Additionally, BASE II will not edit or validate the payment account reference field and will forward the field with the value that was received.

Consumer Presented Quick Response Code Token Processing

The Visa Token Service supports token transactions submitted with a quick response (QR) code.

Acquirers that choose to submit payment token transactions with a QR code must be able to support the values for QR code in V.I.P. 0100/0110 Authorization request and response and 0200/0210 Full financial request and response POS authorization full chip messages.

Acquirers must be able to support chip data, Visa Contactless, and Field 55—Integrated Circuit Card (ICC)-Related Data or expanded third bitmap.

Acquirers that choose to submit token QR code transactions must ensure that their merchants' terminals can support QR readers.

Note:

Currently, QR code transactions are available in the U.S. and A.P. regions only.

Quick Response Code V.I.P. Processing for VTS

Quick Response (QR) code values exist for V.I.P. Token processing.

These values are:

- **03** (Optical code) which may be present in Field 22—Point-of-Service Entry Mode Code
- **3** (Optical code) which may be present in Field 60.2—Terminal Entry Capability

Quick Response Code BASE II Processing for VTS

Quick Response (QR) code values exist for BASE II Draft Data, TCx5 and TCx6, TCR 0.

These values are:

- **3** (Optical code) which may be present in position 158, POS Terminal Capability
- **03** (Optical code) which may be present in positions 162–163, POS Entry Mode

Edit Package Token Processing

The account range definition (ARDEF) table contains all valid ARDEF entries, namely the account prefix range, its associated BIN, card number length indicator, check digit indicator, product ID, account funding source, and an indicator that identifies account ranges that participate in the Visa Token Service.

Acquirers and acquirer processors are reminded to ensure that all platforms are processing based on the processing and routing attributes defined for each nine-digit account range record definition and routing tables.

Acquirers participating in the Visa Token Service can use the Edit Package ARDEF table to identify token account ranges as defined by the Token Indicator, Space (Not applicable), or **Y** (Token range), and can be found in these files and positions:

- ARDEF Value Table Detail Record, position 54
- ARDEF Extract File, position 34
- ARDEF Table Override Part 2, position 33

PC Edit Package for Windows Token Processing

The PC Edit Package also defines the Token Indicator field with the same values used to indicate token ranges, including Space (Not applicable) or **Y** (Token).

Token Edit Package Reports

Acquirers that participate in the Visa Token Service must support the token indicator field in certain Edit Package reports.

These reports are:

- EP-302 Account Range Table Report (Formatted)
- EP-302 Account Range Table Report (Dump)

These optional transaction based reports reflect the transaction layout and values, including token assurance method and PAN token, for the outgoing and incoming files:

- EP-100A Validation Exception Report (Detail)
- EP-204A Returned Item Report
- EP-206A CRS Returned Item
- EP-704 Reclassification Advice

- EP-7x5 Sales Drafts
- EP-7x6 Credit Vouchers

Exception Item Token Processing

The token is a retain and return value for exception processing. Issuers must retain and return the cardholder PAN in the Account Number field and token values in the token fields, submit them in dispute financials and reversals, and request for copies for tokenized transactions.

Acquirers must be prepared to receive the token PAN in the Account Number field in these exception items, and return the token PAN in representments.

Visa Resolve Online Token Processing

Visa Resolve Online (VROL), enhanced for the Visa Claims Resolution (VCR) initiative, supports the Visa Token Service and acquirers can use the token instead of the card/account number (cardholder PAN) in dispute processing for tokenized transactions.

Acquirers can provide the token in the existing card/account number fields in case actions uploaded through the user interface (UI), Bulk Systems Interface (BSI), or Real-Time Systems Interface (RTSI), and can receive the token in the existing card/account number fields in downloaded case actions.

> **Note:**
>
> For the Bulk System Interface files DM5 and DM7, the absence of the designated token field is controlled by the check box in the member profile entitled Retain Existing Behavior for Card Number/Token for Acquirers.

SMS Raw Data and Reports Token Processing

Acquirers that subscribe to SMS Raw Data, Versions 2.2 and support payment token processing must support the SMS Raw Data V22201 record and fields.

The token will be present in the existing SMS Raw Data records where the primary account number is defined. The token will be present in primary account number fields on the acquirer SMS reports.

This table contains the SMS Raw Data records with PAN or token information.

SMS Raw Data and SMS Reports

Record	Field	Position	Format	Length
V22200—Financial Transaction Record	Card Number	97–115	AN	19
V22201—Financial Transaction Record	PAN Token	7–25		
V22300—Financial Maintenance Transaction Record	Card Number	41–59		
V22310—File Maintenance–Address Verification File Update	Card Number	41–59		
V22400—Administrative Message 1 Transaction Record	Card Number	89–107		
V22600—Advice Notification Transaction Record/Interlink-Specific	Card Number	90–108		
V23200—Financial Transaction Record	Card Number	64–82		

TC 33 Acquirer Capture File Token Processing

The TC 33 Enhanced Acquirer Capture File (application code **CPnn**) includes the transaction data necessary to facilitate the acquirers' end-of-day clearing and settlement processing.

Acquirers use the TC 33 Enhanced Acquirer Capture Files to create clearing transactions for both Visa and non-Visa gateway transactions. The file will contain the token instead of the cardholder PAN when token processing was performed.

Acquirers can choose to subscribe to the TC 33 Enhanced Acquirer Capture File using the Visa Open File Delivery (Visa OFD) Service or the Extended Access (EA) Server.

> **Note:**
>
> Acquirers that choose to subscribe to the TC 33 enhanced Acquirer Capture File for the first time will require full implementation activities and should contact their regional client support representative.

TC 33 Merchant Capture File Token Processing

The TC 33 Merchant Capture File allows participating MDEX endpoints to send end-of-day transaction data to Visa. Visa then converts the data into the TC 33 Acquirer Capture File that will be sent to participating acquirers for the MDEX endpoint.

Acquirers then use this information to facilitate their end-of-day clearing processing with Visa and other payment networks. Acquirers use the TC 33 Acquirer Capture Files to create clearing transactions for both Visa and non-Visa gateway transactions. The file will contain the token instead of the cardholder PAN when token processing was performed.

TC 33–Authorization Records and POSA File Token Processing

The TC 33—Authorization Records (POS and PSR) and the Visa Point-of-Sale Authorization (POSA) File are electronically transmitted data files that include V.I.P. authorization data. Acquirers may use this data when submitting BASE II clearing transactions to Visa. The files will contain the token instead of the cardholder PAN when token processing was performed.

Acquirers can choose to subscribe to the TC 33 Authorization Records and the POSA file, and will be required to support these payment token processing fields:

- TC 33—Authorization Records:
  - TCR 1—Authorization and Incremental Authorization (POS), Additional Information record
  - TCR 1—Authorization Full and Partial Reversal (PSR), Additional Information record
- Point-of-Sale Authorization (POSA) File

U.S. Routing Table Token Processing

Visa provides subscription-based U.S. Routing Tables to acquirers and acquirer processors to identify the acceptance programs associated with issuer card programs and identify token ranges.

The Token Indicator in position 30, containing the values of Space (Not applicable) or **Y** (Token) occurs in each logical record of the U.S. Routing Table Detail Record. Acquirers can choose to subscribe to these routing files and categories as part of their support requirements for processing Visa Token Service token payments:

- Consolidated Routing File
- Consolidated Routing File (no surcharge)
- Interlink and Visa Routing File
- PIN at Point-of-Sale Routing File
- POS Debit Device Routing File
- Visa Routing File

  Note:

  U.S. Routing Tables are available to all acquirers globally and are not intended to be used only by U.S. acquirers.

Acquirers that choose to subscribe to the U.S. Routing Tables for the first time must contact their regional client support representative to initiate implementation activities.

The Visa Open File Delivery (Visa OFD) Service and File Exchange Service (VFES) connections are available as options to receive U.S. Routing Tables.

Chapter 4

Visa Token Service Merchant-Initiated Transactions

Merchant-initiated transactions apply to both PAN-based and token transactions. All acquirers must support the processing rules defined for merchant-initiated transactions.

The Visa payment system was originally designed around the principle that all transactions are initiated based on the instruction of the cardholder. However, with the evolution of commerce, business models have evolved, making it possible for a merchant to initiate a transaction based on a prior instruction from the cardholder. As a result, Visa has defined two classes of transactions:

1. Cardholder-initiated transactions
2. Merchant-initiated transactions

Transparency is an important principle in the processing of transactions. It allows all stakeholders within the payment value chain to understand the nature and risk of a specific transaction.

Enhanced processing of incremental authorizations provides more data that can help to improve the accuracy of cardholder funds hold management and increases approval rates.

Merchant-initiated transaction processing addresses:

- Framework and rules for merchant-initiated token transactions, such as tokenized mobile wallet transactions at hotels
- Better approval decisions with insights into the purpose of a transaction
- Better ability to link related transactions together

Merchant-initiated transactions are organized into:

- Industry-specific business practices
- Standing instructions

Industry-Specific Business Practices for VTS Merchant-Initiated Transactions

Transactions under this category are performed to fulfill a business practice as a follow-up to an original cardholder merchant interaction that could not be completed with a single transaction.

These transactions include:

- Incremental authorization transactions
- Resubmission transactions
- Reauthorization transactions
- Delayed charges transactions
- No Show transactions

Incremental Authorization Transactions for VTS

Incremental authorizations can be used to increase the total amount authorized if the authorized amount is insufficient. An incremental authorization request may also be based on a revised estimate of what the cardholder may spend.

Incremental authorizations do not replace the original authorization. They are additional to previously authorized amounts. The sum of all linked estimated and incremental authorizations represent the total amount authorized for a given transaction.

An incremental authorization must be preceded by an estimated/initial authorization. One or more incremental authorizations can be requested while the transaction has not yet been finalized (submitted for clearing). Incremental authorizations must not be used once the original transaction has been submitted for clearing. Instead, a new authorization must be requested, with the appropriate reason code, such as delayed charges or reauthorization.

U.S. CPS/Retail Credit, Debit, and Prepaid payment services support the requested payment service (RPS) value of **I** (CPS/Retail—Incremental Authorizations) with extended clearing timeframes for transactions from these MCCs:

- **4457** (Boat rentals and leasing)
- **7033** (Trailer parks and campgrounds)
- **7394** (Equipment, tool, furniture, and appliance rental and leasing)
- **7519** (Motor home and recreational vehicle rentals)
- **7999** (Recreation services (not elsewhere classified))

Transactions with MCC **5812** (Eating places and restaurants) may qualify for U.S. CPS/Restaurant when utilizing incremental authorizations.

Transactions with MCC **5813** (Drinking places (alcoholic beverages)–bars, taverns, night clubs, cocktail lounges, discotheques) and **7996** (Amusement parks, circuses, carnivals, and fortune tellers) may qualify for CPS/Retail Credit, Debit, and Prepaid payment services with incremental authorizations. MCCs **5813** and **7996** are exempt from the amount tolerance check that requires the authorization amount to exactly match the clearing amount for CPS/Retail Debit and Prepaid.

> **Note:**
>
> For more information, refer to the *U.S. Interchange Reimbursement Fee Rate Qualification Guide*.

These V.I.P. Systems messages are impacted for incremental authorizations:

- 0100/0120 Authorization Request and STIP Advice
- 0400/0420 Reversal Request and Reversal Advice
- 0400/0420 Partial Reversal Request and Partial Reversal Advice

Non-CPS Endpoints

Non-CPS acquirers must submit incremental authorization transactions with all the existing requirements for card-present and card-not-present transactions of the type submitted.

CPS Endpoints

CPS acquirers must support all the existing requirements for CPS incremental authorization transactions. Additionally, CPS acquirers may optionally send Field 63.3—Message Reason Code, populated with the value of **3900** (Incremental authorization) for incremental authorization transactions.

When a transaction identifier is present in both Field 62.2 and TLV Field 125, Usage 2, the value in TLV Field 125, Usage 2 is used in V.I.P. for processing merchant-initiated transactions.

Note:

For incremental authorization transactions, when the acquirer provides the original or previous transaction identifier, V.I.P. will not assign a new one. As a result, the same transaction identifier will be populated in Field 62.2 and TLV Field 125, Usage 2, Dataset ID 03, Tag 03.

Related concepts

MCCs for Incremental Authorization for VTS

Resubmission Transactions for VTS

This is an event that occurs when the original purchase occurred, but the merchant was unable to obtain authorization at the time the goods or services were provided.

A resubmission is only valid when the original authorization presented was declined for insufficient funds. There are a limited number of merchant categories approved to utilize this type of merchant-initiated transaction, and a resubmission is only valid for 14 days after the cardholder’s purchase.

A merchant performs a resubmission in cases where it requested an authorization, but received a decline due to insufficient funds after it has already delivered the goods or services to the cardholder. Merchants in such scenarios can resubmit the request to recover outstanding debt from cardholders.

Acquirers must submit resubmission transactions with all the existing requirements for a card-not-present transaction of the type submitted. Field 63.3—Message Reason Code must have a value of **3901** (Resubmission).

These V.I.P. Systems messages that are impacted for resubmission transaction processing are:

- 0100/0120 Authorization Request and STIP Advice
- 0200/0220 Full Financial Request and STIP Advice

Important:

Resubmission merchant-initiated transactions submitted with a payment token must contain a transaction identifier that is less than 14 days old. Transactions containing a transaction identifier older than 14 days will be declined by V.I.P. with the existing decline code 05 (Do not honor).

Reauthorization Transactions for VTS

A merchant initiates a reauthorization when the completion or fulfillment of the original order or service extends beyond the authorization validity limit set by Visa.

There are two common reauthorization scenarios:

1. Split or delayed shipments at e-commerce retailers–A split shipment occurs when not all of the goods ordered are available for shipment at the time of purchase. If the fulfillment of the goods takes place after the authorization validity limit set by Visa, e-commerce merchants perform a separate authorization to ensure that cardholder funds are available.
2. Extended stay hotels, car rentals, and cruise lines–A reauthorization is used for stays, voyages, and/or rentals that extend beyond the authorization validity period set by Visa.

Acquirers must submit reauthorization transactions with all the existing requirements for a card-not-present transaction of the type submitted. Field 22—Point-of-Service Entry Mode Code must have a value of **01** (Manual (key entry)) or **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file). Field 63.3—Message Reason Code must have a value of **3903** (Reauthorization).

A reauthorization can be submitted up to 90 days from the original purchase, except for select merchant category codes (MCCs), which can be submitted by a merchant as a reauthorization up to 120 days from the original date of purchase.

All other merchants have 90 days to submit a reauthorization.

These V.I.P. Systems messages are impacted for reauthorization transaction processing:

- 0100/0120 Authorization Request and STIP Advice
- 0200/0220 Full Financial Request and STIP Advice

Related concepts

MCCs for Reauthorization Transactions for VTS

Delayed Charges Transactions for VTS

A delayed charge transaction is performed to process a supplemental account charge after original services have been rendered and respective payment has been processed.

Acquirers must submit delayed charges with all the existing requirements for a card-not-present transaction of the type submitted. Field 22—Point-of-Service Entry Mode Code must have a value of **01** (Manual (key entry)) or **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file). Field 63.3—Message Reason Code must have a value of **3902** (Delayed charges).

These V.I.P. Systems messages are impacted for delayed charge transaction processing:

- 0100/0120 Authorization Request and STIP Advice
- 0200/0220 Full Financial Request and STIP Advice

No Show Transactions for VTS

Cardholders can use their Visa cards to make a guaranteed reservation with certain merchant segments. A guaranteed reservation ensures that the reservation will be honored and allows a merchant to perform a no-show transaction to charge the cardholder a penalty according to the merchant’s cancellation policy.

For merchants that accept token-based payment credentials to guarantee a reservation, it is necessary to perform a cardholder-initiated transaction (Account Verification Service) at the time of reservation to be able perform a no-show transaction later.

Acquirers must submit no show transactions with all the existing requirements for a card-not-present transaction of the type submitted. Field 22—Point-of-Service Entry Mode Code should have a value of **01** (Manual (key entry)) or **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file). Field 63.3—Message Reason Code must have a value of **3904** (No show).

These V.I.P. Systems messages are impacted for no show transaction processing:

- 0100/0120 Authorization Request and STIP Advice
- 0200/0220 Full Financial Request and STIP Advice

Standing Instructions for Initial Merchant-Initiated Transactions

Transactions under this category are provided by the customer in the initial cardholder-initiated transaction.

These transactions include:

- Installment payment transactions
- Recurring payment transactions
- Unscheduled credential on file transactions

Installment Payment Transactions for VTS

An installment payment is a transaction in a series of transactions that uses a stored credential and that represents a cardholder agreement for the merchant to initiate one or more future transactions over a period for a single purchase of goods or services.

Acquirers must submit installment payment transactions subsequent to the original installment transaction with either Field 62.2—Transaction Identifier (Bitmap Format) or Field 125—Usage 2: Supporting Information (TLV Format) with the original transaction identifier, and all the existing requirements for a card-not-present transaction of the type submitted.

Field 22—Point-of-Service Entry Mode Code should have a value of **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file). Field 126.13—POS Environment must have a value of **I** (Indicates the message is for an installment payment).

These V.I.P. Systems messages are impacted for installment payment transaction processing:

- 0100/0120 Authorization Request and STIP Advice
- 0200/0220 Full Financial Request and STIP Advice

Recurring Payment Transactions for VTS

A recurring payment transaction is a transaction in a series of transactions that uses a stored credential that is processed at fixed, regular intervals (not to exceed one year between transactions), representing a cardholder agreement for the merchant to initiate future transactions for the purchase of goods or services provided at regular intervals.

Acquirers must submit recurring payment transactions subsequent to the original recurring transaction with either Field 62.2—Transaction Identifier (Bitmap Format) or Field 125, Usage 2: Supporting Information (TLV Format) with the original transaction identifier, and all the existing requirements for a card-not-present transaction of the type submitted.

Field 22—Point-of-Service Entry Mode Code should have a value of **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file). Field 126.13—POS Environment must have a value of **R** (Indicates that the cardholder and merchant have agreed to periodic billing for goods and services, such as utility bills and magazines).

Note:

Field 63.3—Message Reason Code is not used to identify recurring transactions.

These V.I.P. Systems messages are impacted for recurring payment transaction processing:

- 0100/0120 Authorization Request and STIP Advice
- 0200/0220 Full Financial Request and STIP Advice

Unscheduled Credential on File Transactions for VTS

An unscheduled credential on file transaction is a transaction using a stored credential for a fixed or variable amount that does not occur on a scheduled or regularly occurring transaction date, where the cardholder has provided consent for the merchant to initiate one or more future transactions.

Acquirers must submit Field 126.13—POS Environment with a value of **C** (Credential on file) and either Field 62.2 or TLV Field 125—Usage 2 with the original transaction identifier, along with all existing requirements for a card-not-present transaction of the type submitted. Field 22—Point-of-Service Entry Mode Code should have a value of **10** (Credential on file: Merchant initiates transaction for cardholder using credentials stored on file).

> **Note:**
>
> Field 63.3—Message Reason Code is not used to identify unscheduled stored credential transactions.

These V.I.P. Systems messages are impacted for unscheduled and stored credential transaction processing:

- 0100/0120 Authorization Request and STIP Advice
- 0200/0220 Full Financial Request and STIP Advice

Processing Impacts for VTS

Acquirers that support the Visa Token Service and process card-not-present transactions must support the field usage and values to successfully process merchant-initiated transactions with payment tokens.

There are key fields related to merchant-initiated transaction processing relating to:

- V.I.P.
- BASE II Draft Data
- SMS Raw Data and Reports
- Exception Item Processing

V.I.P. Processing Impacts for VTS

Acquirers that support the Visa Token Service and process card-not-present transactions must support V.I.P. field usage and values to successfully process merchant-initiated transactions with payment tokens.

Field 62.1

Field 62.1–Authorization Characteristics Indicator (Bitmap Format), with the value of **I** (Increment to previously approved transaction), applicable to the U.S.

Field 62.2

Field 62.2—Transaction Identifier (Bitmap Format) will support the transaction identifier value for incremental authorization transactions.

Acquirers submitting merchant-initiated transactions must send the original transaction identifier value in Field 62.2 when it is not sent in Field 125, Usage 2, Supporting Information (TLV Format). Acquirers sending the original transaction identifier value in TLV Field 125, Usage 2 are not required to send Field 62.2 in the merchant-initiated transaction request message.

To assist with merchant readiness in supporting the merchant-initiated transaction framework when required, Visa has provided the Visa acquirer-assigned interim transaction identifier to use in place of an original transaction identifier by their merchants on an interim basis. This identifier should be populated in either Field 62.2 (Bitmap format) or TLV Field 125, Usage 2, Dataset ID 03. Acquirers that require the Visa acquirer-assigned interim transaction identifier value must contact their regional client support representative. (Acquirers must not send the value of **0100000000000000** (Issuer interim identifier).)

U.S. CPS incremental authorization request messages may be sent without either Field 62.2 or TLV Field 125, Usage 2.

Field 63.3

Field 63.3—Message Reason Code will contain a value to identify the merchant-initiated transaction type.

Acquirers must send Field 63.3 with one of the merchant-initiated message reason code values in the request message for applicable merchant-initiated transactions. Additionally, acquirers must be aware that V.I.P. will reject any original authorization and purchase request messages sent with Field 63.3, populated with any value other than one of the merchant-initiated message reason codes with the existing reject code **0114** (Invalid value). These are the valid message reason codes for a merchant-initiated transaction:

- **3900** (Incremental authorization)
- **3901** (Resubmission)
- **3902** (Delayed charges)
- **3903** (Reauthorization)
- **3904** (No show)

  Note:

  Field 63.3 is optional in U.S. CPS incremental authorization request messages and is not sent in installment and recurring request messages where Field 126.13 is used to identify the transaction as installment or recurring. This is mandatory for payment token transactions, and optional for cardholder PAN transactions.

Field 125, Usage 2

Field 125—Usage 2: Supporting Information (TLV Format), Dataset 3, Tag 03—Original Transaction Identifier will carry the transaction identifier of the original authorization in a series of merchant-initiated transactions.

Acquirers submitting merchant-initiated transactions must send the original transaction identifier value in TLV Field 125, Usage 2, Dataset ID 03—Additional Original Data Elements, Tag 03—Original Transaction Identifier when it is not sent in Field 62.2. Acquirers sending the original transaction identifier value in Field 62.2 are not required to send TLV Field 125, Usage 2 in the merchant-initiated transaction request message.

To assist with merchant readiness in supporting the merchant-initiated transaction framework when required, Visa has provided the Visa acquirer-assigned interim transaction identifier to use in place of an original transaction identifier by their merchants on an interim basis. this identifier should be populated in either Field 62.2 (Bitmap format) or TLV Field 125, Usage 2, Dataset ID 03. Acquirers that require the Visa acquirer-assigned interim transaction identifier value must contact their regional client support representative. (Acquirers must not send the value of **0100000000000000** (Issuer interim identifier).)

Note:

U.S. CPS incremental authorization request messages may be sent without either Field 62.2 or TLV Field 125, Usage 2. The changes are: Conditional for payment token transactions and Conditional for cardholder PAN transactions.

Field 126.13

Field 126.13—POS Environment will contain one of these values, as applicable:

- **C** (Credential on file) to identify transactions where customer credentials are placed on file for the first time and the new merchant-initiated transaction type of unscheduled stored credential transaction,
- **I** (Indicates that the message is for an installment payment), or
- **R** (Indicates that the cardholder and merchant have agreed to periodic billing for goods and services, such as utility bills and magazines) for payment token recurring transactions

Acquirers must send the value of **C** (Credential on file) in cardholder-initiated transactions when a merchant has placed the cardholder credentials on file for the first time. For installment and recurring transactions, Field 126.13 must be sent with the existing values of **I** (Indicates that the message is for an installment payment) or **R** (Indicates that the cardholder and merchant have agreed to periodic billing for goods and services, such as utility bills and magazines) respectively in cardholder-initiated transactions.

Note:

U.S. CPS installment and recurring request messages, in addition to the new requirement to send Field 126.13, must continue to send Field 60.8—Mail/Phone/Electronic Commerce and Payment Indicator or Field 63.6, position 4—Mail/Phone/Electronic Commerce (MOTO/ECI) and Payment Indicator to meet CPS qualification requirements.

BASE II Draft Data, TCx5 and TCx6 Processing Impacts for VTS

Certain BASE II field usage and values are used in merchant-initiated transactions.

TCR 1, position 168, POS Environment will contain the value of **C** (Credential on file), **I** (Indicates that the message is for an installment payment), or **R** (Recurring payment) for acquirers to identify transactions where customer credentials are placed on file for the first time, and the new merchant-initiated transaction type of unscheduled stored credential transaction.

SMS Raw Data and Reports for VTS

Certain key fields are present in SMS Raw Data and Reports records.

Field 22—Point of Service Entry Mode Code will be present in:

- V22220 record, in the POS Entry Mode field, positions 9–11
- V23202 record, in the POS Entry Mode field, positions 9–11

Field 63.3—Message Reason Code that is used to identify merchant-initiated transactions will be present in:

- V22210 record, in the Message Reason Code field, positions 50–53
- V22400 record, in the Message Reason Code field, positions 85–88
- V23200 record, in the Message Reason Code field, positions 60–63

Field 126.13—POS Environment will be present in the V22220 record in the Recurring Payment Indicator Flag field, position 130.

Exception Item Processing for VTS

Acquirers must retain and return key fields in disputes for subsequent use in dispute responses.

Reversals must be done using the same reversal processing rules used for cardholder-initiated transactions, where the key values used to identify the transaction being reversed are those from the merchant-initiated transactions as documented in the VisaNet technical specifications. Field 63.3—Message Reason Code must be populated with one the valid values for reversal transactions, not the message reason code used in the merchant-initiated transaction.

Related Information

For more information on reversal message requirements, refer to:

- *VisaNet Authorization-Only Online Message Technical Specifications*
- *SMS POS (Visa & Visa Electron) Technical Specifications*
- *Merchant-Initiated Transactions Framework Guide*

Chapter 5

Client-Attended Testing

Acquirers are required to successfully complete attended testing with the Visa Global Client Testing team to participate in the Visa Token Service. Visa provides testing assistance as part of the Visa Token Service. Attended testing is typically performed utilizing the end-to-end payment flow.

For more information on the Visa testing process and requirements, refer to the client versions of :

- *VCMS V.I.P. Testing Guide*
- *VCMS Testing Guide—BASE II*
- *VisaNet Testing Guide*

Appendix A

BASE II Draft Data Records for Visa Token Service

Certain BASE II fields and positions are required to support the Visa Token Service for acquirers. These fields and values are applicable to all token types.

Only the token data fields for BASE II Draft Data are included.

BASE II Draft Data Transactions for Visa Token Service – TC x5, TC x6, TC x7

Token transactions are valid for certain transaction code qualifier (TCQ) values.

These TCQ values are:

- **0** (Default)
- **1** (Account funding)
- **2** (Original credit)

TCR 0 Record Token Data Fields for BASE II Draft Data

This table shows the layout and contents of the token data fields of the BASE II Draft Data, TCR 0 record.

TCR 0

Position	Length	Format	Field Name	Description
1–4	4	n/a	n/a	n/a
5–20	16	UN	Account Number	This field contains the cardholder PAN. For token transactions, this field will contain the token.
21–151	131	n/a	n/a	n/a

TCR 0

Position	Length	Format	Field Name	Description
152–157	6	AN	Authorization Code	This field contains the authorization code. A token transaction must be authorized and this field must contain the same value as the authorization request. If an invalid value is submitted, the transaction will be returned by BASE II with return reason code **17** (The authorization code is invalid) or rejected by the Edit Package with reject code **V0195** (The authorization code is invalid). Invalid authorization codes are: • **00000** (All zeros in the last five positions of the field) • **^^^^^** (All spaces in the last five positions of the field) • **X** (**X** in the last position of the field) • **0000^** (Four zeros followed by a space in the last five positions of the field) • **0000N** (Four zeros followed by **N** in the last five positions of the field) • **0000Y** (Four zeros followed by **Y** in the last five positions of the field) • **0000P** (Four zeros followed by **P** in the last five positions of the field) • **SVC** (**SVC** in the first three positions of the field)
158	1	AN	POS Terminal Capability	This field contains the capability of the POS terminal.
159–161	3	n/a	n/a	n/a

TCR 0

Position	Length	Format	Field Name	Description
162–163	2	AN	POS Entry Mode	This field contains the POS entry mode. This field should contain the same POS entry mode value as the authorization request.
164–168	5	n/a	n/a	n/a

TCR 1 - Additional Data Token Data Fields for BASE II Draft Data

This table shows the layout and contents of the token data fields of the BASE II Draft Data, TCR 1 record.

TCR 1 - Additional Data

Position	Length	Format	Field Name	Description
1–4	4	n/a	n/a	n/a
5	1	AN	Business Format Code	This field is reserved for future use. This field contains a space.
6–7	2	AN	Token Assurance Method	A value that allows the Token Service provider to indicate the trust level of the Token to PAN/ cardholder. It is determined as a result of the type of identification and verification (ID & V) performed and the entity that performed it. If not used, this field should be space filled. **Note:** This edit is applicable for the TC 05 (POS originals) and TC 25 (reversals).
8–168	113	n/a	n/a	n/a

TCR 1 - Additional Data

Position	Length	Format	Field Name	Description
122	1	AN	Additional Token Response Information	This field contains a value that identifies if the transaction is eligible for token services. Valid token service values are: • **1** (Token program) • Space (Not applicable) **Note:** Endpoints must support receiving any alphanumeric value in this field.
123–168	45	n/a	n/a	n/a

TCR 4 - Supplemental Financial Data Token Data Fields for BASE II Draft Data

This table shows the layout and contents of the token data fields of the BASE II Draft Data, TCR 4—Supplemental Financial Data record.

TCR 4 - Supplemental Financial Data

Position	Length	Format	Field Name	Description
1–119	4	n/a	n/a	n/a
120–148	29	AN	Payment Account Reference	This field will contain the payment account reference.
149–159	11	AN	Token Requestor ID	This field will contain the token requestor ID. If not used, this field should be space filled.
160–168	9	n/a	n/a	n/a

TCR 4 - Supplemental Financial and Promotion Data Token Data Fields for BASE II Draft Data

This table shows the layout and contents of the token data fields of the BASE II Draft Data, TCR 4—Supplemental Financial and Promotion Data record.

TCR 4 - Supplemental Financial Data

Position	Length	Format	Field Name	Description
1–111	111	n/a	n/a	n/a
112–140	29	AN	Payment Account Reference	This field will contain the payment account reference.
141–151	11	AN	Token Requestor ID	This field will contain the token requestor ID. If not used, this field should be space filled.
152–168	17	n/a	n/a	n/a

TC 40 - Fraud Advice Transaction Layout Token Data Fields for BASE II Draft Data

This table shows the layout and contents of the payment account reference (PAR) in the BASE II TC 40 fraud advice transaction.

TCR 2 Record

Position	Length	Format	Field Name	Description
1–87	87	n/a	n/a	n/a
88–116	29	AN	Payment Account Reference	This field contains the payment account reference.
117–168	52	n/a	n/a	n/a

TC 52 - Retrieval Request Record Transaction Layout Token Data Fields for BASE II Draft Data

This table shows the layout and contents of the token data fields in the BASE II TC 52 for transactions destined to acquirers.

TCR 0 Record

Position	Length	Format	Field Name	Description
1–4	4	n/a	n/a	n/a
5–20	16	UN	Account Number	This field contains the cardholder PAN. For token transactions, this field will contain the token.
21–168	148	n/a	n/a	n/a

Appendix B

BASE II TC 50 Text Message Transactions for Commercial Card for VTS

Acquirers must be aware of the BASE II fields and positions in the various types of the TC 50 Free Text Messages that support the Visa Token Service for commercial card data.

BASE II Free Text Messages - TC 50 (Formatted) for Commercial Card for VTS

In the commercial card TC 50 free text formatted messages, the token should be placed in the account number field. There are no other token-specific fields in the TC 50s.

TCR 0 Information for TC 50 (Formatted) for Commercial Card for VTS

These are the formatted commercial card TC 50 types and the associated service identifiers, as well as the account number positions.

TCR 0

TC 50 Type	Service Identifier (positions 17–22)	Account Number Positions	Account Number Extension
Passenger Itinerary Data	CORPAI COMMAG	42–57	58–60
Generic Data	COMMGN	42–57	58–60
Invoice (Header and Summary)	PURCHA COMMXA	121–136	137–139
Lodging (Additional Data Elements)	CORPLA COMMLA	121–136	137–139
Car Rental (Additional Data Elements)	CORPCD COMMCA	121–136	137–139

BASE II Free Text Messages - TC 50 (Open Format, Fixed Length) for Commercial Card for VTS

In the commercial card TC 50 open format, fixed length text messages, the token should be placed in the account number field. In these TC 50 messages, the merchant type is identified by the merchant sector value.

References

For more information, see the *Visa U.S.A. Enhanced Data Service Member Implementation Guide*. For a current copy of the guide, contact Visa U.S.A. Commercial Solutions.

TCR 0 Information for TC 50 (Open Format, Fixed Length) for Commercial Card for VTS

These are the open format fixed length commercial card TC 50 types and the merchant sector values, as well as the account number positions.

TCR 0

TC 50 Type	Service Identifier (positions 17–22)	Merchant Sector Value (positions 38–39)	Account Number Positions	Account Number Extension
Open Format	OPNFMT	SC (Shipping and overnight courier services) TS (Temporary help services)	47–62	63–65

BASE II Free Text Messages - TC 50 (XML Format) for Commercial Card for VTS

The commercial card TC 50 XML format varies from the fixed length and open format. This TC 50 is an open format with the first 46 positions formatted and the token should still be placed in the account number field; however, the account number may be anywhere within the TC 50, and is identified by the element <CardNum>.

References

For more information, see:

- *Visa U.S.A. Enhanced Data Service Member Implementation Guide*. For a current copy of the guide, contact Visa U.S.A. Commercial Solutions.
- *Visa XML Invoice Implementation Guide*

TCRs 0-7 for TC 50 (XML Format) for Commercial Card for VTS

This table lists the location of the <CardNum> element.

XML Data contained in the TC 50

Section	Element Name
Invoice Summary/ActualPayment/Cardinfo	<CardNum>

Appendix C

Applicable Commercial Card Products for Tokenization

Acquirers must be aware of which Commercial Card products are eligible for tokenization.

Commercial Card Products Eligible for Tokenization

This table shows the eligible commercial card products for tokenization.

Applicable Commercial Card Products for Tokenization

Commercial Product	Product Name	Product ID	Product Subtype
Business Card	Visa Business	**G^**	n/a
	Visa Signature Business	**G1**	n/a
	Visa Platinum Business	**G3**	n/a
	Visa Infinite Business	**G4**	n/a

Applicable Commercial Card Products for Tokenization

Commercial Product	Product Name	Product ID	Product Subtype
Business Card (AP region)	Visa Business Rewards	**G5**	n/a
Corporate Card	Visa Corporate	**K^**	n/a
	Visa Government Corporate T&E	**K1**	n/a
Purchasing Card	Visa Purchasing	**S^**	n/a
	Visa Purchasing with Fleet	**S1**	n/a
	Visa Government Purchasing	**S2**	n/a
	Visa Government Purchasing with Fleet	**S3**	n/a
Purchasing Card (LAC region)	Visa Commercial Agriculture	**S4**	n/a
	Visa Commercial Transport/Cargo	**S5**	n/a
	Visa Commercial Marketplace	**S6**	n/a
	Visa Commercial Distribution	**S^**	**DS**
	Visa Commercial Construction	**S^**	**CS**
	Visa Commercial Healthcare	**S^**	**HC**
Purchasing Card (AP region)	Visa Commercial Distribution	**S^**	**DS**

Appendix D

TC 33 Merchant Capture Files for VTS

Participating acquirers must be aware of the fields and positions in the TC 33.B—Merchant Capture File that support the Visa Token Service. These fields and values are applicable to all token types.

TC 33.B Merchant Capture Files for VTS

The TC 33 Merchant Capture File is an electronically transmitted data file that includes the transaction data necessary for clearing.

The data is available to acquirers that choose to subscribe to the TC 33 records.

References

For more information on the TC 33 merchant capture files, see the *BASE II Clearing Interchange Formats, TC 01 to TC 49*.

TC 33.B - Merchant Capture File – CP51 TCR 0 - Transaction Data (for VTS)

This table shows the layout of the TC 33.B, CP51 TCR 0.

TC 33.B, CP51 TCR 0 - Transaction Data Record

Position	Length	Format	Field Name	Description
1–64	64	n/a	n/a	n/a
65–80	16	AN	Account Number	This field will contain either the PAN, or the token for the PAN.
81–83	3	AN	Account Number Extension	This field will contain the PAN extension, the PAN token extension, or all spaces.
84–87	4	AN	Expiration Date	This field will contain the cardholder PAN expiration date, or the token expiration date in mmyy format, where: • mm=month • yy=year
88–161	74	n/a	n/a	n/a
162–165	4	AN	Message Reason Code	This field will contain one of the following valid values: • **3900** (Incremental authorization) • **3901** (Resubmission) • **3902** (Ancillary charges) • **3903** (Reauthorization) • **3904** (No show)
166–168	3	n/a	n/a	n/a

TC 33.B - Merchant Capture File – CP51 TCR 1 - Additional Data (for VTS)

This table shows the layout of the TC 33.B, CP51 TCR 1.

TC 33.B, CP51 TCR 1 - Additional Data Record

Position	Length	Format	Field Name	Description
1–117	117	n/a	n/a	n/a
118	1	AN	CAVV Results Code	This field will contain the cardholder authentication verification value (CAVV) results code that identifies the outcome of the CAVV validation.
119–170	52	n/a	n/a	n/a

TC 33.B - Merchant Capture File – TCR 8 - Supplemental Data (for VTS)

The following table shows the layout of the TC 33.B, CP51 TCR 8.

TC 33.B, CP51 TCR 8 - Supplemental Data Record

Position	Length	Format	Field Name	Description
1–40	40	n/a	n/a	n/a
41–42	2	AN	Token Assurance Method	For transactions that contain token-specific data, this field will contain the token assurance value that indicates the confidence level of the token to PAN mapping. For all transactions that do not contain token-specific data, this field will contain all spaces.
43–53	11	UN	Token Requestor ID	This field will contain the token requestor ID.
54–72	19	AN	PAN Account Range	This field will contain a variable length value of 0 to 19 digits. The value can be either the first nine digits of the PAN account range, or up to the full cardholder PAN.

TC 33.B, CP51 TCR 8 - Supplemental Data Record

Position	Length	Format	Field Name	Description
73	1	AN	Regulated/ Non-Regulated Status	This field will contain one of the following values to indicate the status of the account range: • **R** (Regulated) • **N** (Non-regulated)
74–80	7	n/a	n/a	n/a
81–109	29	AN	Payment Account Reference Number (PAR)	This field will contain the payment account reference value, generated by VisaNet.
110–170	61	n/a	n/a	n/a

Appendix E

TC 33 Acquirer Capture Files for VTS

Participating acquirers must be aware of the fields and positions in the TC 33.A Enhanced Acquirer Capture File (**CPnn** in the Application Code field) that support the Visa Token Service. These fields and values are applicable to all token types.

TC 33.A Acquirer Capture File for VTS

The TC 33.A Acquirer Capture File is an electronically transmitted data file that includes the transaction data necessary for clearing.

The data is available to acquirers that choose to subscribe to the TC 33 records.

TC 33.A, CP01 TCR 0-Transaction Data (Enhanced) for VTS

This table shows the layout of the enhanced TC 33.A, CP01 TCR 0.

TC 33.A, CP01 TCR 0-Transaction Data Record

Position	Length	Format	Field Name	Description
1–62	62	n/a	n/a	n/a
63–78	16	AN	Account Number	This field will contain either the PAN, or the token for the PAN.
79–81	3	AN	Account Number Extension	This field will contain the PAN extension, the PAN token extension, or all spaces.
82–85	4	AN	Expiration Date	This field will contain the cardholder PAN expiration date, or the token expiration date in mmyy format, where: • mm=month • yy=year
86–159	74	n/a	n/a	n/a
160–163	4	AN	Message Reason Code	This field will contain of the following valid values: • **3900** (Incremental authorization) • **3901** (Resubmission) • **3902** (Ancillary charges) • **3903** (Reauthorization) • **3904** (No show)
164–168	5	n/a	n/a	n/a

TC 33.A, CP01 TCR 1-Additional Data (Enhanced) for VTS

This table shows the layout of the enhanced TC 33.A, CP01 TCR 1.

TC 33.A, CP01 TCR 1-Additional Data

Position	Length	Format	Field Name	Description
1–167	167	n/a	n/a	n/a
168	1	AN	CAVV Results Code	This field will contain the cardholder authentication verification value (CAVV) results code that identifies the outcome of the CAVV validation.

TC 33.A, CP01 TCR 8-Supplemental Data (Enhanced) for VTS

This table shows the layout of the enhanced TC 33.A, CP01 TCR 8.

TC 33.A, CP01 TCR 8-Supplemental Data

Position	Length	Format	Field Name	Description
1–38	38	n/a	n/a	n/a
39–40	2	AN	Token Assurance Method	This field contains the Token Assurance Method values.
41–51	11	UN	Token Requestor ID	This field will contain the token requestor ID.
52–70	19	AN	PAN Account Range	This field will contain a variable length value of 0 to 19 digits. The value can be either the first nine digits of the PAN account range, or up to the full cardholder PAN.
71	1	AN	Regulated/ Non-Regulated Status	This field will contain one of the following values to indicate the status of the account range: • **R** (Regulated) • **N** (Non-regulated)
72–78	7	n/a	n/a	n/a
79–107	29	AN	Payment Account Reference Number (PAR)	This field will contain the payment account reference value, generated by VisaNet.

TC 33.A, CP01 TCR 8-Supplemental Data

Position	Length	Format	Field Name	Description
108–161	53	n/a	n/a	n/a
162	1	AN	Additional Token Response Information	This field contains a value that identifies if the transaction is eligible for token services. Valid token service values are: • **1** (Token program) • Space (Not applicable) **Note:** Endpoints must support receiving any alphanumeric value in this field.
163–168	5	n/a	n/a	n/a

Appendix F

Authorization Records–BASE II TC 33 (POS, PSR) and the Point-of-Sale Authorization (POSA) File for VTS

Participating acquirers must be aware of the fields and positions in the TC 33 Authorization Records (POS and PSR) and the POSA File that supports the Visa Token Service. These fields and values are applicable to all token types.

BASE II TC 33 Records and the POSA File for VTS

The TC 33 Authorization Records and the Visa POSA File are electronically transmitted data files that include the V.I.P. authorization data. Acquirers can use this data when submitting BASE II clearing transactions to Visa.

The data is available to subscribing acquirers.

TC 33-Authorization and Incremental Authorization (POS) Record Layout for VTS

This table shows the layout of the TC 33, TCR 0.

TC 33, TCR 0-Authorization and Incremental Authorization (POS) Record Layout

Position	Length	Format	Field Name	Description
1–33	33	n/a	n/a	n/a
34	1	AN	Additional Token Response Information	This field contains a value that identifies if the transaction is eligible for token services. Valid token service values are: • **1** (Token program) • Space (Not applicable) **Note:** Endpoints must support receiving any alphanumeric value in this field.
35–82	47	n/a	n/a	n/a
83–98	16	AN	Account Number	When a token is present in the 0100/0110 Authorization request message, this field will contain the token. When the token is not present in the 0100/0110 Authorization request message, this field will contain the cardholder primary account number (PAN).
99–114	16	n/a	n/a	n/a
115–118	4	AN	Expiration Date	When a token is present in the 0100/0110 Authorization request message, this field will contain the expiration date for the token. When the token is not present in the 0100/0110 Authorization request message, this field will contain the expiration date for the cardholder PAN in mmyy format, where: • mm=month • yy=year
119–168	50	n/a	n/a	n/a

This table shows the layout of the TC 33, TCR 1.

TC 33, TCR 1-Authorization and Incremental Authorization (POS), Additional Information Record Layout

Position	Length	Format	Field Name	Description
1–142	142	n/a	n/a	n/a
143	1	AN	Regulated Account Status	This field will contain one of the following values to indicate the status of the account range: • **R** (Regulated) • **N** (Non-regulated) **Note:** All transactions without a token will contain the value **N** (Non-regulated).
144–152	9	UN	Primary Account Number, Account Range	When a token is present in the 0100/0110 Authorization request message, this field will contain the first nine digits of the cardholder PAN. When a token is not present in the 0100/0110 Authorization message, this field will contain all **zeros**. Acquirers should be aware that the first nine digits of the cardholder PAN must not be forwarded to their merchants.
153–154	2	AN	Token Assurance Method	When a token is present in the 0100/0110 Authorization request message, this field contains token assurance method value. When a token in not present in the 0100/0110 Authorization request message, this field contains all spaces.
155–165	11	UN	Token Requestor ID	This field will contain the token requestor ID. When a token is present in the 0100/0110 Authorization request message, this field will contain the token requestor ID value. When a token is not present in the 0100/0110 Authorization message, this field will contain all **zeros**.
166–168	3	n/a	n/a	n/a

This table shows the layout of the TC 33, TCR 4.

TC 33, TCR 4, Authorization and Incremental Authorization (POS), Additional information Record Layout

Position	Length	Format	Field Name	Description
1–4	4	n/a	n/a	n/a
5–33	29	AN	Payment Account Reference	This field will contain the payment account reference value.
34–168	135	n/a	n/a	n/a

TC 33-Authorization Full and Partial Reversal (PSR) Record Layout for VTS

This table shows the layout of the TC 33, TCR 0.

TC 33, TCR 0-Authorization Full and Partial Reversal (PSR) Record Layout

Position	Length	Format	Field Name	Description
1–33	33	n/a	n/a	n/a
34	1	AN	Additional Token Response Information	This field contains a value that identifies if the transaction is eligible for token services. Valid token service values are: • **1** (Token program) • Space (Not applicable) **Note:** Endpoints must support receiving any alphanumeric value in this field.
35–80	45	n/a	n/a	n/a
81–96	16	AN	Account Number	When a token is present in the 0100/0110 Authorization request message, this field will contain the token. When the token is not present in the 0100/0110 Authorization request message, this field will contain the cardholder primary account number (PAN).
97–112	16	n/a	n/a	n/a

TC 33, TCR 0-Authorization Full and Partial Reversal (PSR) Record Layout

Position	Length	Format	Field Name	Description
113–116	4	AN	Expiration Date	When a token is present in the 0100/0110 Authorization request message, this field will contain the expiration date for the token. When the token is not present in the 0100/0110 Authorization request message, this field will contain the expiration date for the cardholder PAN in mmyy format, where: • mm=month • yy=year
117–168	52	n/a	n/a	n/a

This table shows the layout of the TC 33, TCR 1.

TC 33, TCR 1-Authorization Full and Partial Reversal (PSR) Additional Information Record Layout

Position	Length	Format	Field Name	Description
1–142	142	n/a	n/a	n/a
143	1	AN	Regulated Account Status	This field will contain one of the following values to indicate the status of the account range: • **R** (Regulated) • **N** (Non-regulated) **Note:** All transactions without a token will contain the value **N** (Non-regulated)
144–152	9	UN	Primary Account Number, Account Range	When a token is present in the 0100/0110 Authorization request message, this field will contain the first nine digits of the cardholder PAN. When a token is not present in the 0100/0110 Authorization message, this field will contain all **zeros**. Acquirers should be aware that the first nine digits of the cardholder PAN must not be forwarded to their merchants.

TC 33, TCR 1-Authorization Full and Partial Reversal (PSR) Additional Information Record Layout

153–154	2	AN	Token Assurance Method	When a token is present in the 0100/0110 Authorization request message, this field contains token assurance method value. When a token in not present in the 0100/0110 Authorization request message, this field contains all spaces.
155–165	11	UN	Token Requestor ID	This field will contain the token requestor ID. When a token is present in the 0100/0110 Authorization request message, this field will contain the token requestor ID value. When a token is not present in the 0100/0110 Authorization message, this field will contain all **zeros**.
166–168	3	n/a	n/a	n/a

This table shows the layout of the TC 33, TCR 4.

TC 33, TCR 4, Authorization and Incremental Authorization (POS), Additional information Record Layout

Position	Length	Format	Field Name	Description
1–4	4	n/a	n/a	n/a
5–33	29	AN	Payment Account Reference	This field will contain the payment account reference value.
34–168	135	n/a	n/a	n/a

Point-of-Sale Authorization (POSA) File for VTS

When a token is present in the 0100/0110 Authorization request message, the token will be added in the primary account number field, and the token expiration date in the card expiration date field.

This table shows the layout of the POSA File detail record.

POSA File Detail Record Layout

Position	Length	Format	Field Name	Description
1–63	63	n/a	n/a	n/a
64–91	28	UN	Primary Account Number	This field will contain the PAN or token.
92–105	14	n/a	n/a	n/a
106–109	4	AN	Card Expiration Date	This field will contain the PAN or token expiration date.
110–600	491	n/a	n/a	n/a

Appendix G

SMS Raw Data Record for Token Data Elements

SMS Raw Data V22201 Record for VTS

The SMS Raw Data record V22201 contains the token data elements. The V22201 record will be systematically delivered when an acquirer subscribes to SMS Raw Data, Version 2.2, supports TLV Field 123, Usage 2, Dataset ID 68—Token Data, and key token data elements are present.

SMS Raw Data V22201 Record Layout for VTS

This table shows the V22201 record layout of the token data elements.

SMS Raw Data V22201 Record Layout

Position	Length	Format	Field Name	Description
1–6	6	n/a	n/a	n/a
7–25	19	UN	PAN Token	This field will contain the token that was present in the 0100/0110 Acquirer token request message and will be space filled for all other transactions containing tokens. (TLV Field 123, Usage 2, Dataset ID 68, Tag 01.) **Note:** Acquirers will receive the token in the existing SMS Raw Data records where the primary account number is currently defined today.
26–27	2	AN	Token Assurance Method	This field will contain a value that indicates the confidence level of the token to PAN mapping. (Field 123, Usage 2, Dataset ID 68, Tag 02.)
28–38	11	UN	Token Requestor ID	This field will contain the token requestor ID, if present. If not present, this field will be zero filled. (Field 123, Usage 2, Dataset ID 68, Tag 03.)
39–130	92	n/a	n/a	n/a

SMS Raw Data V22261 Record

The SMS Raw Data record V22261 contains the additional token response information for endpoints that subscribe to SMS Raw Data, Version 2.2 and Version 2.3.

SMS Raw Data V22261 Record Layout for VTS

This table shows the V22261 record layout of the token data elements.

SMS Raw Data V22261 Record Layout

Position	Length	Format	Field Name	Description
1–45	45	n/a	n/a	n/a
46	1	AN	Additional Token Response Information	This field contains a value that identifies if the transaction is eligible for token services. Valid token service values are: • **1** (Token program) • Space (Not applicable) **Note:** Endpoints must support receiving any alphanumeric value in this field.
47–130	92	n/a	n/a	n/a

SMS Raw Data V23210 Record

The SMS Raw Data record V23210 contains the additional token response information for endpoints that subscribe to SMS Raw Data, Version 2.2 and Version 2.3.

SMS Raw Data V23210 Record Layout

This table shows the V23210 record layout of the token data elements.

SMS Raw Data V23210 Record Layout

Position	Length	Format	Field Name	Description
1–30	30	n/a	n/a	n/a
31	1	AN	Additional Token Response Information	This field contains a value that identifies if the transaction is eligible for token services. Valid token service values are: • **1** (Token program) • Space (Not applicable) **Note:** Endpoints must support receiving any alphanumeric value in this field.
32–130	99	n/a	n/a	n/a

Appendix H

MCCs for Incremental Authorization for VTS

MCCs for Incremental Authorization for VTS

Purchase transactions at all MCCs, excluding the transactions shown in the table below, are eligible for incremental authorization transactions.

This table shows the transactions that are excluded from incremental authorization transactions.

Transactions Excluded from Incremental Authorization

Exclusions	
Account funding transactions	Advance payments transactions
ATM cash disbursement transactions	Cryptocurrency transactions
Installment transactions	Manual cash disbursement transactions
Quasi-cash transactions	Recurring transactions

Appendix I

MCCs for Reauthorization Transactions for VTS

MCCs for Reauthorization Transactions for VTS

Participating acquirers must be aware of MCCs eligible for the 120-day time limit check that applies to reauthorization merchant-initiated transactions submitted with a token.

This table lists the valid MCCs for reauthorization merchant-initiated authorization transactions.

MCCs for Reauthorization Transactions

MCC	Description
3351–3441	Car rental agencies
3501–3999	Lodging – hotels, motels, resorts
4411	Steamship and cruise lines
4457	Boat rentals and leasing

MCCs for Reauthorization Transactions

MCC	Description
7011	Lodging – hotels, motels, resorts, central reservation services (not elsewhere classified)
7033	Trailer parks and campgrounds
7394	Equipment, tool, furniture, and appliance rental and leasing
7512	Automobile rental agency
7513	Truck and utility trailer rentals
7519	Motor home and recreational vehicle rentals
7999	Recreation services (not elsewhere classified)

Appendix J

About the Visa Token Services Acquirers Guide–Technical Specifications

The *Visa Token Service Acquirers Guide - Technical Specifications* provides information related on technical requirements, message specifications, and testing for the Visa Token Service. The guide references unique token requirements, not business-as-usual everyday processing.

This guide provides technical details required for acquirers to implement and support token payment transactions in the VisaNet System. Acquirer processors should use this guide in conjunction with the supporting related documents for online processing, clearing and settlement, reporting, and backoffice processing. For a list of documents, refer to Visa Token Service Related Publications later in this chapter.

This guide documents changes up through the April 2023 and July 2023 Business Enhancements Release.

Summary of Impacts for VTS Acquirers Guide–Technical Specifications

The technical, implementation, and testing information provided in this document consists of changes up through the April 2023 and July 2023 Business Enhancements Release.

Summary of Changes for VTS Acquirers Guide–Technical Specifications

Since the *Visa Token Service Technical Specifications Guide for Acquirers, Version 2.0* was published, technical changes have occurred that need to be incorporated to the document. Changes are marked with change bars. The changes in this version are related to articles from the *VisaNet Business Enhancements Global Technical Letter and Implementation Guide*s shown here, and are listed in the Visa Token Service Related Publications section.

- April 2020 and July 2020
- October 2020 and January 2021
- April 2021 and July 2021
- October 2021 and January 2022
- April 2022 and July 2022
- October 2022 and January 2023
- April 2023 and July 2023

Audience for VTS Acquirers Guide–Technical Specifications

The technical information and details in this guide are intended for acquirers and acquirer processors that are managing and executing the implementation and system development tasks for the participation requirements in transaction, record, and file processing for the Visa Token Service.

Region-specific business requirements and participation requirements may vary and may not be called out in this guide. For further information, contact your regional client support representative.

> **Note:**
>
> Within Europe, the Visa Token Service can support any Visa-approved network processor providing authorization, clearing, and settlement. This document uses Visa processing for illustration purposes. For questions about the Visa Token Service in Europe contact your Europe relationship manager.

References

For more information about the tokenization requirements for acquirers in the CEMEA region, refer to CEMEA Clients Should Prepare to Support Tokenization in the *Visa Business News*, dated 7 June 2018.

Table Legends for VTS Acquirers Guide–Technical Specifications

There are specific codes for V.I.P. message formats, and format keys for V.I.P. and BASE II fields and positions used throughout the *Visa Token Service Acquirers Guide-Technical Specifications*.

The V.I.P. Message Format Table Legend table lists the codes used in V.I.P. message layout tables.

V.I.P. Message Format Table Legend

Code	Description
C	Conditional—The field or value is present under certain conditions.
C+	The field or value is conditionally added by V.I.P. at the VIC.
C–	The field or value is conditionally removed by V.I.P. at the VIC.
M	Mandatory—The field or value must be present in the message.
M+	The field or value is always added by V.I.P. at the VIC.
O	Optional—The presence of the field or value in the message is up to the message initiator of the recipient.
Blank	The field or value must not be present in the message in that stage of its journey.

V.I.P. Message Format Table Legend

Code	Description
>	V.I.P. passes the field or value with the message; no V.I.P. action other than possible field editing.
–	The field or value is always removed by V.I.P. at the VIC.

The General Table Legend table lists format keys used in tables that provide field and position information, and in message and record layouts.

General Table Legend

Code	Description
A	Alphabetic
AN	Alphanumeric allowed, no special characters
ANS	Alphanumeric allowed, including special characters
ASCII	American standard code for information interchange
B	Binary value
BCD	Numeric, four-bit BCD = Unsigned packed
DX	Display hexadecimal
EBCDIC	Extended binary coded decimal interchange code
N	Numeric
UN	Unpacked numeric

Related Publications for VTS Acquirers Guide–Technical Specifications

Additional technical and system processing information related to the Visa Token Service are available. The audience for these documents includes acquirers and vendors.

This table shows the related documents for acquirers participating in the Visa Token Service.

Visa Token Service Related Documents

Title and Description	Primary Audience
Visa Token Service manuals: • *EMV Payment Tokenization Specification—Technical Framework* • *Visa Token Service Description* • *Visa Cloud-Based Payments Program Minimum Requirements and Guidelines* • *Visa Cloud-Based Payments Program Description* • *Visa Cloud-Based Payments Contactless Specification* • *Visa Contactless Payment Specification Version 2.0.2, including Additions and Clarifications 3.0 (or higher version)* • *Visa Ready Program for Cloud-Based Payments Program Guide Security Requirements and Evaluation Guidance for Mobile Applications—Visa Digital Solutions EMV®*	Acquirers
V.I.P. System manuals: • *VisaNet Authorization-Only Online Messages – Technical Specifications* • *Full Service POS Online Messages – Technical Specifications* • *V.I.P. System Reports* • *V.I.P. System SMS Processing Specifications*	Acquirers
BASE II System manuals: • *BASE II Clearing Data Codes* • *BASE II Clearing Edit Package Run Control Options Quick Reference* • *BASE II Clearing PC Edit Package for Windows (Release 4.0) User's Guide* • *BASE II Clearing Edit Package (Release 4.0) Messages* • *BASE II Clearing Edit Package (Release 4.0) Operations Guide* • *BASE II Clearing Edit Package (Release 4.0) Reports* • *BASE II Clearing Edit Package (Release 4.0) Run Control Options Quick Reference Guide* • *BASE II Clearing Edit Package, Release 4.0, Migration Implementation Guide* • *BASE II Clearing Interchange Formats, TC 01 to TC 49* • *BASE II Clearing Interchange Formats, TC 50 to TC 92* • *BASE II Clearing Reports* • *BASE II Clearing Services*	Acquirers
Visa Core Rules and Visa Product and Service Rules documentation: • *Visa Rules* • *Visa Product Brand Standards* • *Payment Technology Standards Manual*	Acquirers

Visa Token Service Related Documents

Title and Description	Primary Audience
Business Release documentation: • *April 2014 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 1.1, Mandatory Changes for Visa Resolve Online – Article 2.7, Changes to the Edit Package ARDEF Table – Article 2.15, Changes to BASE II and Edit Package – Article 3.3, Changes to U.S. Routing Tables – Article 3.6, New SMS Raw Data Record for Token Data Elements – Article 3.9, Changes to the TC 33 Acquirer Capture File – Article 3.10, Authorization Records and the POSA File – Article 4.1, New Visa Network Token Service – Article 4.2, Optional Changes for Visa Resolve Online – Article 9.5.1, Requirements for Commercial Card Data in TC 50 Records – Article 11.1, U.S. Mandate for Support of Payment Token Standard Fields – Appendix B, BASE I and V.I.P. Token Messages • *October 2014 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 2.6, Changes to the ARDEF Table – Article 2.7, New Authorization Characteristics Indicator Values in Existing BASE I and V.I.P. Fields – Article 3.1, Changes to Support the Payment Token Framework Fields and Visa Network Token Service for Interlink Messages – Article 3.6, Changes to the Visa Payment Token Framework – Article 3.10, Changes to the TC 33 Acquirer Capture File – Article 4.2, Optional Changes for Visa Resolve Online – Article 5.3, Introduction of the Payment Token Framework Fields in AP, Canada, CEMEA, and LAC – Article 9.4.1, New Visa Europe Payment Token Service for Acquirers – Article 9.5.1, Payment Token Processing Mandate for Visa Europe Acquirers – Article 11.2.4, U.S. Domestic Interchange Fee Program for Application-Based E-Commerce Transactions From Mobile Devices – Article 11.3.1, U.S. Mandate for Support of Payment Token Fields in Interlink Messages • *April 2015 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 3.9, Changes to SMS Raw Data and Reports – Article 4.2, Visa Token Service for Acquirers – Article 4.3, Optional Changes for Visa Resolve Online	Acquirers

Visa Token Service Related Documents

Title and Description	Primary Audience
• *October 2015 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 3.4, Visa Token Service for Cloud-Based Payment Token Transactions with Magnetic Secure Transmission for Acquirers – Article 3.16, Changes to the Visa Token Service for Card-on-File, E-Commerce, and Application-Based E-Commerce Transactions for Acquirers – Article 9.5.1, Payment Token Processing for Visa Europe Acquirers • *April 2016 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 2.2, Changes to Enforce Authorization Code Processing for Payment Token Transactions – Article 3.1, Visa Token Service Commercial Transaction Processing for Acquirers – Article 3.7, Changes to Support Interregional Application-Based E-Commerce Transactions from Secure Element Mobile Devices – Article 5.11, Introduction of the Payment Account Reference for the Visa Token Service – Article 5.12, Changes to Cloud-Based Payment Token Transactions with Magnetic Secure Transmission – Article 9.5.2, Changes to Commercial Cards for the Visa Europe Payment Token Service • *October 2016 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 1.2, Alignment of Merchant-Initiated Transactions and Authorization Enhancements – Article 2.7, Changes to Edit Package – Article 3.1, New Payment Account Reference Number Field in Visa Token Service Messages and BASE I and V.I.P. Messages – Article 3.10, Token Account Range Assignment to In-Use Primary Account Number Ranges – Article 3.11, Changes to the Enhanced TC 33 Acquirer Capture File – Article 3.12, Changes to the TC 33 Merchant Capture File – Article 3.14, New Cryptogram Authentication for the Visa Token Service to Support Secure Element Mobile Devices – Article 5.1, Introduction of the Payment Account Reference for the Visa Token Service – Article 5.11, Mail/Phone/Electronic Commerce and Payment Indicator Values in E-Commerce and Application-Based E-Commerce Transactions for Acquirers – Article 5.14, Support Card-Present Data for High-Value Transactions – Article 5.16, Expansion of Incremental Authorization Processing – Article 5.20, Mandate for Eight-Digit Issuer BIN	Acquirers

Visa Token Service Related Documents

Title and Description	Primary Audience
• *April 2017 and July 2017 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 2.6, Expansion of Incremental Authorization Processing – Article 2.14, Changes to Add token Requestor ID – Article 2.16, Changes to Support Payment Account Reference in BASE II Draft Data Transactions – Article 3.1, Changes to the Visa Token Service – Article 3.9, Changes to the Enhanced TC 33 Acquirer Capture File – Article 3.10, Changes to the TC 33 Merchant Capture File – Article 5.5, Merchant-Initiated Transaction Framework – Article 5.6, Changes to Incremental Authorization Processing for T&E and Transit Merchants – Article 10.2.1, Support of Payment Token Data Fields for Issuers in LAC – Article 11.2.6, Changes to U.S. CPS Programs to Support Transactions with Incremental Authorizations – Article 11.4.1, Changes to the Visa Token Service POS Transactions with Visa Quick Response Code • *October 2017 and January 2018 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 2.11, Mandate to Support Stored Credential Transaction Framework – Article 2.14, Changes to Payment Account Reference – Article 3.2, Changes to the Visa Token Service to Support Account Funding and Original Credit Transactions – Article 3.14, Changes to 3-D Secure Indicator Field and Clarification on Processing Rules – Article 4.3, Changes to the Visa Token Service to Support Token Transactions with 3-D Secure – Article 9.1.1, Mandate to Support the Merchant-Initiated Transaction Framework in the Europe Region – Article 9.2.1, Changes to Support the Token Requestor ID – Article 9.3.6, Changes to the Visa Token Service to Support Original Credit Transactions in the Europe Region – Article 11.1.1, Requirement for U.S. Issuers to Receive the POS Environment Field – Article 11.4.1, Changes to the Visa Token Service POS Transaction With Visa Quick Response Code – Article 11.4.2, Changes to the Visa Token Service to Support Token Transaction With 3-D Secure in the U.S. Region – Article 11.5.1, Changes to the Visa Token Service to Support Dynamic Token Verification Value in the U.S. Region	Acquirers

Visa Token Service Related Documents

Title and Description	Primary Audience
• *April 2018 and July 2018 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – No acquirer token specific articles • *October 2018 and January 2019 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*: – Article 3.6, Mandate for the Visa Token Service That Supports Token Transactions with 3-D Secure – Article 3.13, Support of the Visa Token Service for Account Funding and Original Credit Transactions – Article 3.17, Changes to the Enhanced TC 33 Acquirer Capture File – Article 3.18, Changes to the TC 33 Merchant Capture File – Article 3.20, Changes to Support BASE II-Originated Original Credit Transactions for the Visa Token Service – Article 4.9, Changes to the Visa Token Service to Support Contactless ATM Transactions – Article 4.10, New V.I.P. Payment Account Reference Inquiry Messages – Article 5.6, Reminder for Acquirers to Deliver Payment Account Reference to Merchants – Article 9.2.2, Changes to Support Payment Account Reference	Acquirers

Visa Token Service Related Documents

<table>
<tr><th>Title and Description</th><th>Primary Audience</th></tr>
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April 2019 and July 2019 VisaNet Business Enhancements Global Technical Letter and Implementation Guide:

Article 3.2, Requirement to Support 3-D Secure
Article 4.16, Changes to Support a New Version of CAVV Data
Article 6.4.1, Expansion of the Visa Token Service POS Transactions with Visa Quick Response Code
Article 8.5.1, Requirements for Enabling Digital Commerce for Acquirers and Issuers in the CEMEA Region
Article 9.2.2, Changes to Support Payment Account Reference

October 2019 and January 2020 VisaNet Business Enhancements Global Technical Letter and Implementation Guide:

Article 2.17, Changes to Original Transaction Identifier Field for Merchant-Initiated Transactions
Article 3.9, Changes to the Primary Account Number in TLV Field 123
Article 5.11, Expansion of Estimated, Initial, and Incremental Authorization Processing for Certain Merchants
Article 5.12, Requirements for Token Authorization Verification Value and Dynamic Token Verification Value for Acquirers and Merchants
Article 6.4.1, Expansion of the Visa Token Service POS Transactions with Visa Quick Response Code
Article 9.1.2, Changes to Support Visa Delegated Authentication
Article 9.2.1, Changes to Support Payment Account Reference

April 2020 VisaNet Business Enhancements Global Technical Letter and Implementation Guide

Article 3.3, Changes to the Visa Token Service to support contactless manual cash disbursement transactions
Article 9.3.2, Changes to the Primary Account Number in TLV Field 123

October 2020 VisaNet Business Enhancements Global Technical Letter and Implementation Guide

Article 5.7, New Deferred Authorization Message Reason Code for Token Transactions

</td><td>Acquirers</td></tr>
</table>

Visa Token Service Related Documents

Title and Description	Primary Audience
• *April 2021 VisaNet Business Enhancements Global Technical Letter and Implementation Guide* – No acquirer token specific articles • *October 2021 VisaNet Business Enhancements Global Technical Letter and Implementation Guide* – Article 2.10, Changes to Support the Visa Secure Credential Framework – Article 2.11, Changes to Support Token Assurance Method – Article 3.12, Changes to Processing for a Token Return Reason Code in BASE II – Article 4.1, Option to Support Visa Secure Credential Framework – Article 11.2.5, Changes in Authorization Characteristics Indicator Processing for Token-Based E-Commerce Transactions • *April 2022 VisaNet Business Enhancements Global Technical Letter and Implementation Guide*' – Article 1.1, Support of Visa Secure Credential Framework – Article 3.13, Changes to Support Token Solutions for Web Browser Auto Fill – Article 5.9, Visa Documentation Revised to Include Credential Update Terminology – Article 11.5.1, Changes in Authorization Characteristics Indicator Processing for Token-Based E-Commerce Transactions • *October 2022 VisaNet Business Enhancements Global Technical Letter and Implementation Guide* – Article 2.2, Changes to Support Token Solutions for Web Browser Auto Fill – Article 3.3, Changes to Token Life Cycle Management for Suspending and Resuming Tokens – Article 3.5, Changes to Token Authentication for Account Funding Transactions – Article 11.5.1, Changes in Authorization Characteristics Indicator Processing for Token-Based E-Commerce Transactions • *April 2023 VisaNet Business Enhancements Global Technical Letter and Implementation Guide* – Article 11.2.2, Changes in Authorization Characteristics Indicator Processing for Token-Based E-Commerce Transactions	Acquirers

Glossary

Application-Based E-Commerce	Includes secure element and host card emulation.
Card-on-File	This is most applicable to merchants who store user credentials, including the PAN.
E-Commerce	A platform that allows sales and purchases of goods and services over an electronic network – typically the internet.
Host Card Emulation (HCE)	HCE enables mobile applications running on supported operating systems with the ability to offer payment card and access card solutions independently of third parties while leveraging cryptographic processes traditionally used by hardware-based secure elements without the need for a physical secure element.
Magnetic Secure Transmission (MST)	Simulates a physical magnetic-stripe transaction sent from a mobile phone or other device.
Near Field Communication (NFC)	A set of wireless technologies that use a magnetic field to facilitate communication between devices when they are within a specified radius from each other.
Payment Account Reference	A data element in authorization messages and clearing transactions that is associated with a cardholder's primary account number (PAN). When the PAN is tokenized, the payment account reference will then be associated with all the payment tokens for the cardholder PAN.
POS Entry Mode	A value that is used to indicate the token presentment mode through which the token is presented for payment. Each network will define and publish any new POS entry mode values as part of its existing message specifications.

Token
A surrogate value for a PAN that is consistent with ISO 8583 message requirements, and is a 13 to 19-digit numeric value that must pass basic validation rules of an account number, including the Luhn formula, also known as the mod 10 algorithm. Payment tokens are generated within a BIN range that has been designated as a token BIN range and flagged accordingly in all appropriate BIN tables. Payment tokens must not have the same value as or conflict with a cardholder PAN.

Token Assurance Method
A value that allows the Token Service Provider to indicate the confidence level of the payment token to PAN/cardholder binding. It is determined as a result of the type of identification and verification (ID&V) performed and the entity that performed it. The token assurance method is set when provisioning a payment token and may be updated if additional ID&V is performed. The token assurance method is defined by the Token Service Provider.

Token Cryptogram
A cryptogram that is uniquely generated by the token requestor through cryptography to validate authorized use of the token. The cryptogram will be carried in different fields in the transaction message based on the type of transaction and associated use case.

Token Expiration Date
The expiration date of the token that is generated by and maintained in the token vault and passed in the PAN Expiration Date field during transaction processing to ensure interoperability and minimize the impact of tokenization implementation.

Token Requestor ID
A value that uniquely identifies the pairing of token requestor with the token domain. Thus, if a given token requestor needs token for multiple domains, it will have multiple token requestor IDs, one for each domain. It is assigned by the Token Service Provider and is unique within the token vault.

Visa Contactless
Tap and pay-enabled contactless cards.